EPGP-404 · ENVIRONMENTAL GOVERNANCE & SUSTAINABILITY

Easy-to-Understand Study Notes

Every topic from all seven class documents, explained in plain language with an everyday example for every topic, and each numerical problem solved step by step using the method shown in class.

How to read these notes

EXPLAINED SIMPLY

Every topic opens with a short, plain explanation and an everyday example (many from India), before the formal class content.

IN SIMPLE WORDS

A plain-language explanation of the idea, often with an everyday analogy.

REMEMBER FOR THE EXAM

Definitions, frameworks and lists you should be able to reproduce.

CASE / EXAMPLE

The real-world cases used in class and what each one teaches.

NUMERICAL – SOLVED

Worked problems, using the professor's formulas and steps.

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1. **The Sustainability Imperative**
   Lubin & Esty (HBR 2010): sustainability as a business megatrend, the 4 stages of value creation, the 5 execution areas, and Losers/Defenders/Dreamers/Winners
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   Pelenc, Ballet & Dedeurwaerdere (UN GSDR brief 2015): substitutability, critical natural capital, comparison table
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   4 capitals, Hartwick–Solow rule, K_{crit}, nested systems, planetary boundaries, 3 market failures, double materiality, Rio → Kyoto → Paris → CBAM, **CBAM steel numerical**
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   Role of prices, why pricing fails, Wunder's 5 criteria, WTP/WTA, **upstream–downstream PES game numerical**, tank-water PES, Deepwater Horizon
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   Hardin's pasture model (**herdsman numerical**), goods matrix, property regimes, bundle of rights, Ostrom's SES & 8 design principles, **Hardin vs Ostrom game numerical**
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MODULE 1 · CLASS I READING

The Sustainability Imperative

Source: David A. Lubin & Daniel C. Esty, Harvard Business Review, May 2010 (Reprint R1005A)

1.1 The big idea in one paragraph

EXPLAINED SIMPLY

Every few decades something comes along that changes the rules of business for *everyone*. The authors say sustainability is one of these waves. Companies that learn to ride it will grow, and those that ignore it will shrink.

> **Example:** When UPI arrived in India, shops that started accepting QR payments kept their customers, and those that insisted on cash only lost sales. Nobody could opt out. Sustainability works the same way.

Business history has long calm periods broken by big shifts that change how every company competes. Electricity, mass production, globalisation, the quality movement and information technology (IT) were all shifts of this kind. The authors call them **business megatrends**. They argue that **sustainability is the next megatrend**. Firms that study how winners handled earlier megatrends can see a clear road map for this one.

IN SIMPLE WORDS

Think of the arrival of smartphones. Companies that adapted early (apps, mobile banking) thrived, and those that ignored it (Nokia, BlackBerry) faded. The authors say environmental pressure is the same kind of wave. You cannot opt out of it; you can only ride it well or badly.

REMEMBER FOR THE EXAM: THE ARTICLE IN THREE LINES

1. **The Big Idea:** Sustainability is an emerging business megatrend, like electrification and mass production. It will affect firms' competitiveness and even their survival.
2. **The argument:** Studying how leaders won earlier megatrends, especially the **quality movement** and the **rise of IT**, helps firms build a strategy for this one.
3. **The better approach:** (a) build a **vision** that moves through **four stages of value creation**, defensive first and offensive later; (b) build **execution capabilities in five areas**: leadership, assessment (methods), strategy development, management integration, and reporting & communication.

1.2 What is a "megatrend"?

EXPLAINED SIMPLY

A megatrend is a big, long-lasting shift that changes **how companies compete**. It is not a fashion that passes in a year.

> **Example:** Electricity was a megatrend: every factory had to rebuild around electric motors. The internet was another. Today, rules on carbon, water and waste are forcing every firm to redesign how it works.

John Naisbitt popularised the word in his 1982 book *Megatrends*, which was about broad social shifts such as globalisation, the information society and the move from hierarchies to networks. Lubin and Esty narrow it to **business megatrends**: *fundamental and persistent shifts in how companies compete*.

- They come from **technological innovation** or **new ways of doing business**.
- Financial crises, social shifts, or conflict over resources can start or speed them up. For example, the Cold War drove the space race and microelectronics, which led to the IT megatrend.
- Examples: electrification, mass production, globalisation, the quality movement (1970s–80s), IT.
- **Common thread:** each one gave corporate leaders a **strategic imperative they could not escape**.

1.3 Why sustainability qualifies as a megatrend

EXPLAINED SIMPLY

Several forces are pushing at once: scarce resources, governments writing new rules, investors asking about carbon and water, consumers wanting greener products, and huge sums flowing into clean technology. When so many forces point the same way, it is a megatrend and not a fad.

> **Example:** A cement company in India now faces carbon disclosure questions from investors, water limits from the state, CBAM charges if it exports, and buyers asking for 'green cement', all at the same time.

Over the last decade, environmental issues have steadily cut into firms' ability to create value. The drivers:

Driver	What it means
Globalised workforces and supply chains	These created environmental pressures and new business liabilities.
Rise of China and India	More competition for natural resources (especially oil), which adds a geopolitical angle.
Externalities becoming "material"	CO_2 emissions and water use now matter to investors, and stakeholders expect firms to disclose them.
Public and government concern	Climate change, pollution, food safety and resource depletion. Examples: the SEC ruling that climate risk is material to investors, and the EPA mandate to regulate greenhouse gases as a pollutant.
Consumer demand	Consumers seek sustainable products or push firms to improve traditional ones.
Huge investment flows (sidebar "Fueling the Megatrend")	Clean-tech venture investment ran at about **$9 billion a year** in 2008. Private clean-tech investment was estimated at **over $200 billion a year**. G20 governments earmarked about **$400 billion of $2.6 trillion** in stimulus for clean tech and sustainability (≈15%).

Conclusion: managers can no longer ignore sustainability as a central factor in long-term competitiveness.

1.4 Learning from past megatrends: quality and IT

EXPLAINED SIMPLY

We have seen big waves before (quality and IT), and winners followed the same path each time: first save money, then redesign, then grow revenue, then build a brand around it. So we can copy that path for sustainability.

Example: Toyota first used quality to cut defects (save money). It ended up making 'Toyota = reliable' its brand. A firm can likewise start by cutting energy bills and end up being known as 'the green brand'.

In both earlier megatrends, market leaders moved through the **same four stages of value creation**:

1. Cut **cost, risk and waste**, and prove the value ("proof-of-value").
2. **Redesign** selected products, processes or functions, moving from *doing old things in new ways* to *doing new things in new ways*.
3. Drive **revenue growth** by building the innovation into the core strategy.
4. **Differentiate** through new business models, using the innovation to strengthen culture, brand and other intangibles for lasting advantage.

THE QUALITY STORY

- The late-1970s downturn and the 1979 oil shock pushed consumers toward efficiency. The car industry was transformed.
- After WWII, Japan made cheap, low-quality goods, and "Made in Japan" was not a brand asset. By the mid-1970s Japanese leaders had adopted **W. Edwards Deming's** ideas on quality as a core value.
- **Toyota and Honda** adopted **Total Quality Management (TQM)**. This broke the old belief that *high quality and affordability cannot go together*.
- The **Toyota Way** applied quality from concept to customer, and later to brand and culture. Toyota's later recall crisis shows that a newer top-line growth strategy clashed with its long-term quality strategy, so **core strategy elements must stay aligned**.
- **Motorola** (Rey Moore) first used quality to *detect* faults (cut cost, waste, risk), then redesigned processes to *prevent* faults, and then created **Six Sigma** with tools such as Pareto charts and root-cause analysis. Quality became part of its brand and culture.

THE IT STORY

- The 1982 recession pushed firms to use IT for productivity. Early results were mixed, and skeptics called IT a "black hole" for money.
- **American Airlines** captured over **40%** of all US airline transactions with its **Sabre** reservation system.
- **American Hospital Supply (AHS)** built **ASAP** (Analytic Systems Automatic Purchasing), which let hospitals order supplies electronically. AHS went from cutting cost, time and errors, to service innovation ("any product, any manufacturer, any time, any desktop"), to a new business model (running hospitals' inventory and procurement). Earnings grew from **$42 million (1974) to $237 million (1984)**.

SMALL NUMERICAL – AHS EARNINGS GROWTH

1. Growth multiple = 237 ÷ 42 = **5.64 times** in 10 years.
2. Compound annual growth rate (CAGR) = $(237/42)^{1/10} - 1 = 5.64^{0.1} - 1 \approx 1.189 - 1 =$ **≈ 18.9% per year**.

AHS's IT-led strategy grew earnings about 5.6× (≈19% a year) over 1974–84.

TWO PROBLEMS TO SOLVE AT THE SAME TIME

Firms that want advantage in sustainability must **(1) formulate a vision** for value creation ("*what* we must do") and **(2) execute on it** ("*how* we must do it") with new structures, methods, roles and processes.

1.5 Getting the vision right: the four stages of value creation

EXPLAINED SIMPLY

Think of four steps on a staircase: (1) do the same things more efficiently, (2) redesign products and processes, (3) make green products a main source of revenue, (4) build the whole business model and brand around sustainability.

> **Example:** A paint company might: (1) cut energy and waste in its factory, (2) remove toxic lead from all paints, (3) launch a best-selling 'eco paint' range, (4) become known as the healthy-home company and charge a premium.

The idea goes back to Esty & Winston's *Green to Gold* (2006) and was extended by Nidumolu, Prahalad & Rangaswami ("Why Sustainability Is Now the Key Driver of Innovation", HBR 2009).

Stage	What firms do	Example from the article
1. Do old things in new ways (defensive)	Beat competitors on regulatory compliance and on cutting environment-related cost and risk. Build proof that eco-efficiency pays.	**3M's Pollution Prevention Pays (PPP)**, started about 30 years earlier. By 2005 it had cut pollutants by over **2.6 billion pounds** and saved over **$1 billion**. Its 2005–2010 targets cut energy, emissions and waste costs by another **20%**.
2. Do new things in new ways	Redesign products, processes and whole systems across the value chain for resource efficiency and risk management.	**DuPont's "zero waste"** commitment. It sold high-footprint businesses (carpets, nylon) because their business and environmental risks outweighed their future earnings.
3. Transform the core business (offensive)	Sustainability innovation becomes a source of **new revenue and growth**.	**Dow's 2015 Sustainability Goals** led to solar roof shingles and hybrid batteries, shifting Dow from commodity chemicals toward advanced materials and energy.
4. New business model creation and differentiation	Use the megatrend to differentiate the business model, brand, employee engagement and other intangibles, repositioning the whole firm.	**GE ecomagination**: about **$25 billion** revenue expected in 2010. It repositioned GE as an energy and environmental solutions provider with a "green aura".

IN SIMPLE WORDS

Stages 1–2 are **defence**: save money, avoid fines, waste less. Stages 3–4 are **offence**: make money from green products and become known as the green leader. Most firms start with defence and earn the right to play offence.

1.6 Getting execution right: the five capability areas

EXPLAINED SIMPLY

A good vision is useless if you can't deliver it. Execution needs five things: a senior leader in charge (CSO), proper tools to measure, a clear strategy, sustainability built into everyday management and bonuses, and honest reporting.

Example: A firm announces 'zero waste by 2030' (vision). Without a CSO, waste meters, targets in each plant manager's bonus and a yearly report, the promise stays on paper. That is like announcing a diet without a weighing scale or a meal plan.

In each area the firm must move **from tactical, ad hoc and siloed** to **strategic, systematic and integrated**. The sidebar "Capturing the Eco-Premium" summarises the shifts.

Area	From (tactical)	To (strategic)	Examples in the article
1. Leadership ("Elevate Leadership")	Individual, departmental leadership	C-level leadership linking sustainability strategy to initiatives and outcomes. Shared goals with suppliers, customers and stakeholders.	The **Chief Sustainability Officer (CSO)** should become like the CIO: strategic, with resources and responsibilities. **Coca-Cola** worked with bottlers to "light-weight" packaging, saving tens of millions of dollars. It committed to **water neutrality** with WWF, and to **HFC-free** coolers by 2015 after urging from Greenpeace (99% less GHG from that equipment).
2. Methods for assessing value ("Systematize Methods and Models")	Imprecise, inconsistently used tools that track discrete projects	Professional green business analysis with specialised tools (scenario planning, risk modelling), certifications and standards	"If you can't measure it, you can't manage it." **Fujitsu's "cost green index"** scorecard. **3M** applies **lean Six Sigma** to energy, waste and GHG, with **55,000 employees** trained.
3. Strategy development ("Align Strategy and Deployment")	Unconnected tactical programmes	Megatrend-based initiatives aligned with core strategy and built into operating plans	Open-source strategy that involves outsiders. **Wal-Mart Sustainability 360** (2006, CEO Lee Scott): 100% renewable energy, zero waste, lower GHG. A dozen **Sustainable Value Networks** bring together staff, NGOs, academics, officials and suppliers. One network improved fleet efficiency by **38%**, saving **over $200 million a year** and cutting **200,000 tons** of GHG a year.
4. Management integration ("Integrate Management")	Siloed responsibility	Shared accountability through integrated objectives and performance reviews	**Dow** links sustainability to pay and reviews. Newly promoted unit managers must review their unit's sustainability plan with senior management **within 90 days**. **IBM's** environmental management system (Wayne Balta) supports policy, goals, decisions and data from R&D to end-of-life.
5. Reporting & communication ("Systematize Reporting and Communication")	Ad hoc reporting with non-standard tools	Sustainability scorecards, benchmarking, best-practice comparison, consistent internal and external communication	Wall Street will demand evidence of returns. The authors scored firms on **35 dimensions**, and scores from public information alone were almost always lower than scores from full inside information. **Timberland** puts "nutrition-label" style environmental-impact labels on its products.

1.7 Sidebars: the eco-premium and advice for first movers

EXPLAINED SIMPLY

Eco-premium: firms that manage environmental risk well are seen as less risky, so investors lend to them more cheaply and their share price is steadier. **First-mover warning:** being first to go green helps, but competitors catch up, so you must keep innovating.

Example: Two similar companies borrow money. The one with strong pollution controls and a water plan is less likely to face a shutdown or fine, so the bank charges it a slightly lower interest rate. That saving is the eco-premium.

EVIDENCE OF AN ECO-PREMIUM

Studies link strong environmental and sustainability performance with better financial returns. **Julie Fox Gorte's** analysis of a 2009 **Mercer** survey of several dozen studies found that firms with better social and environmental performance tend to have a **lower cost of capital** because their risk is lower. Better environmental risk management also **reduces volatility** in shareholder value and earnings. The authors call this boost in shareholder value the **eco-premium**.

ADVICE FOR FIRST MOVERS: DON'T REST ON YOUR GREEN LAURELS

Early leadership does not guarantee lasting advantage, and continued innovation is needed. **Wang** led office computing but failed to keep up when desktops arrived. Competitors eventually copied **AHS's** innovations, but AHS still delivered superior value for a decade, "not a bad ride".

1.8 The sustainability performance system: the 2×2 matrix

EXPLAINED SIMPLY

Ask two questions about any firm: how big is its green **ambition**, and how strong is its ability to **execute**? The answers place it in one of four boxes: Losers (low/low), Defenders (low ambition, good execution), Dreamers (high ambition, poor execution) and Winners (high/high).

Example: A student who plans to top the class but never studies is a **Dreamer**. One who only aims to pass but studies steadily is a **Defender**. One with neither plan nor effort is a **Loser**. One with a big goal and a solid study plan is a **Winner**.

Vision ("what we must do") plus execution ("how we must do it") make up a firm's **sustainability performance system**. Firms fall into four groups:

	Capacity for tactical execution	Capacity for strategic execution
Value creation from product & business-model transformation (ambitious vision)	**DREAMERS** big vision, weak execution	**WINNERS** big vision, strong execution
Value creation from waste, cost & risk reduction (modest vision)	**LOSERS** modest vision, weak execution	**DEFENDERS** modest vision, strong execution

Category	Meaning	Examples
Losers	Only modest cost/risk/waste efforts and a vague or disjointed vision. They will find it harder and harder to protect their position.	From earlier megatrends: **GM** (failed to see how quality would transform autos) and **Kodak** (missed digital photography).
Defenders	A deliberate "go-slow" strategy focused on early-stage cost, risk and waste. This works *if* the gap with competitors does not grow too large and execution matches the complexity of the business.	**Maersk**: slow-speed shipping cuts fuel cost and CO_2. That is fine as long as rivals don't pursue a more sweeping strategy, such as more efficient ship design.
Dreamers	Vision runs ahead of the ability to execute. Risk of unkept promises, **greenwashing** charges, and reputational and financial harm.	**London Stock Exchange** paperless settlement: budgeted at **£6 million**, abandoned in 1993 after costs passed **£400 million**. **Ford** under Bill Ford: unmet SUV fuel-economy and hybrid promises drew criticism. Successor Alan Mulally delivered efficient models that helped Ford weather the downturn.
Winners	Clear vision plus strong strategic execution.	**GE ecomagination**: tens of billions in revenue, leadership in energy infrastructure, efficient appliances, jet engines and locomotives, plus a "halo effect" on GE's reputation. **Clorox Greenworks**: billions in sales, and the purchase of **Burt's Bees** convinced stakeholders the shift was sincere.

SMALL NUMERICAL – THE "DREAMER" COST OVERRUN (LONDON STOCK EXCHANGE)

1. Planned cost = £6 million. Actual cost by 1993 = over £400 million.
2. Overrun multiple = 400 ÷ 6 ≈ **66.7 times** the budget.
3. Percentage overrun = (400 − 6) ÷ 6 × 100 ≈ **6,567%**.

Lesson: ambition without execution capability can cost about 67× the plan. That is the "Dreamer" trap.

CLOSING MESSAGE IN SIMPLE WORDS

Like quality and IT, sustainability will touch **every function, every business line and every employee**. Scorecards and richer environmental data will let analysts identify firms that can deliver an eco-premium. Firms with a clear vision and real execution capability will come out ahead, and the rest "will be left by the wayside."

MODULE 2 · CLASS I READING

Weak versus Strong Sustainability

Source: Pelenc, Ballet & Dedeurwaerdere, Brief for the UN Global Sustainable Development Report (GSDR) 2015

2.1 The fundamental debate

The core question in sustainable development is whether we adopt a **weak** or a **strong** idea of sustainability. Everything turns on one word: **substitutability**, meaning whether natural capital can be replaced by other forms of capital.

IN SIMPLE WORDS

Picture a family that inherits a forest. **Weak sustainability** says: "Cut it all down, sell the timber and put the money in a bank. As long as total family wealth doesn't fall, we're fine." **Strong sustainability** says: "Some things the forest does (clean water, rain, soil, wildlife) can never be bought back with money. Keep at least the critical part of the forest."

2.2 Weak sustainability

EXPLAINED SIMPLY

Weak sustainability treats all kinds of wealth as interchangeable. If you use up nature but build machines, roads or schools worth the same amount, you are 'sustainable', because the total stays the same.

Example: A village sells its forest for ₹10 crore and builds a ₹10 crore school and road. Under weak sustainability the village is no poorer, since the forest just became a school.

- Assumes **natural capital and manufactured capital are essentially substitutable**, with no essential difference in the well-being they create (Ekins et al. 2003; Neumayer 2003, 2012).
- Only the **total value of the aggregate capital stock** matters. It should be at least **maintained**, ideally increased, for future generations (**Solow, 1993**).
- Famous quote (Neumayer 2003): *"it does not matter whether the current generation uses up non-renewable resources or dumps CO_2 in the atmosphere as long as enough machineries, roads and ports are built in compensation."*
- This leads to **maximising monetary compensation** for environmental damage.
- It assumes **technological progress** will keep producing solutions to environmental problems caused by producing more goods and services.

2.3 Strong sustainability

EXPLAINED SIMPLY

Strong sustainability says some of nature can never be swapped for money or machines. Nature and man-made things **work together** (complements); they don't replace each other (substitutes).

Example: A tractor is useless without fertile soil, and a water purifier is useless without a river. Once the soil or river is gone, you can't buy it back. That is why strong sustainability insists on keeping nature intact.

Natural capital is **not just a stock of resources**. It is a set of **complex, evolving systems** of living (biotic) and non-living (abiotic) elements whose interactions decide how well ecosystems can provide functions and services to society. There are three reasons why natural capital **cannot be fully substituted**:

Reason	Explanation	Layman's example
1. Qualitative difference and irreversibility	Manufactured capital can be reproduced and its destruction is rarely irreversible. Using up natural capital is usually **irreversible** (species extinction). We also lack knowledge of how natural systems work, so we can't be sure what destroying them does to well-being. Irreversibility and uncertainty call for the **precautionary principle**.	A bridge that collapses can be rebuilt. An extinct tiger cannot be brought back.
2. Manufactured capital needs natural capital	Machines are made from nature, so they can never fully replace it. Natural capital's contribution is **multidimensional**. The **Millennium Ecosystem Assessment (2005)** lists four contributions of ecosystem services to well-being: **security, basic materials for a good life, health, and good social relations**. Natural capital is therefore a **complement** to other capitals, not a substitute.	A fishing boat is useless without fish, and the boat itself is made of steel and fuel taken from nature.
3. More future consumption is not fair compensation	Losses of natural capital cannot be paid back with more goods later. UNDP (2011): "Today's generation cannot ask future generations to breathe polluted air in exchange for a greater capacity to produce goods and services." This is an **intergenerational justice** issue.	You can't tell your grandchildren, "Sorry the air is toxic, but here's a bigger TV."

2.4 Critical natural capital

EXPLAINED SIMPLY

Critical natural capital is the 'must-keep' part of nature: the parts we depend on for life itself and which technology can't replace. It must never fall below a safe level.

Example: A city's only freshwater lake, the bees that pollinate local crops, or a mangrove forest protecting a coast from cyclones. Losing them would hurt lives, not just profits.

DEFINITION TO MEMORISE

Critical natural capital is the part (particular configuration) of natural capital that provides ecosystem services **critical for human existence and well-being**. It must be kept above **certain thresholds of degradation** so that nature can keep providing those services (Noël & O'Connor 1998; Ekins et al. 2003; Brand 2009).

- Strong sustainability does **not** say every ecosystem service everywhere must be kept exactly as it is. It says we must identify the ones that are especially important for life and well-being, and target policy at them.
- **The difficulty:** because our knowledge of ecosystems is uncertain, it is very hard to judge which services are critical and which are not.

2.5 The scientific debate: deciding what is "critical"

EXPLAINED SIMPLY

Deciding what is 'critical' isn't purely a science question. Scientists can measure thresholds, but society must also decide what risks it accepts and what it values. That needs public discussion, not just experts.

> **Example:** Should a sacred grove near a village be protected? Ecologists may call it 'not critical' for water, but the community sees it as critical for its culture. Both views count, so the decision should be made together.

- **Brand (2009)** names **six domains** where natural capital can be critical: **socio-cultural, ecological, sustainability, ethical, economic, and human survival**.
- Because natural capital and well-being are both complex and multidimensional, the measures are often **not comparable or commensurable** (Scheidel 2013).
- "Objective" ecological criteria include **safe minimum standards, minimum ecosystem size, maximum sustainable yield, and ecological footprint**. Societal values, ethics and attitudes to risk also shape what counts as critical.
- So defining what is critical needs **both factual science and a normative (value-based) basis**. We must go beyond purely technical, expert calculations to **public deliberation and stakeholder participation** (Van den Hove 2000).
- Implementing strong sustainability needs a **trans-disciplinary approach**. Natural science identifies **ecological thresholds and planetary boundaries**, but this must be combined with social science and a broad societal debate on (i) **acceptable levels of risk**, especially for vulnerable populations, and (ii) the **values behind human development**.

2.6 Comparison table (adapted from Mancebo, 2013)

EXPLAINED SIMPLY

Weak = 'total wealth must not fall; any form is fine'. Strong = 'some natural wealth is irreplaceable; protect it whatever the total'.

Example: If a company drains a lake and builds a bigger factory, weak sustainability asks 'did total value rise?' and might say yes. Strong sustainability asks 'was that lake critical?', and if so the answer is no, whatever the factory is worth.

Aspect	Strong sustainability	Weak sustainability
Key idea	Substituting natural capital with other capital is **severely limited**	Natural and other capitals (manufactured etc.) are **perfectly substitutable**
Consequences	Some human actions can have **irreversible** consequences	**Technological innovation** and **monetary compensation** for environmental damage
Sustainability issue	Conserve the irreplaceable stocks of **critical natural capital** for future generations	The **total value** of the aggregate capital stock should be at least maintained, ideally increased
Key concept	Critical natural capital	Optimal allocation of scarce resources
Defining thresholds and norms	Scientific knowledge as **input to public deliberation** (procedural rationality)	A **technical/scientific** approach sets thresholds (instrumental rationality)

AUTHORS' CONCLUSIONS (ISSUES FOR FURTHER CONSIDERATION)

- Scientifically, **strong sustainability is to be greatly preferred**, because assuming full substitutability of natural capital "appears improbable".
- Improve **multidimensional, integrated assessment** of how nature and human well-being interact, such as integrated assessment of ecosystem services.
- Build a **normative basis** to judge the sustainability of these interactions from a strong-sustainability viewpoint.

MODULE 3 · CLASS 1 SLIDES (22 SEPT 2026)

From Natural Capital to Global Governance

Parts: (1) Natural capital & substitution · (2) Why environment precedes S & G · (3) Need for global governance · (4) Governance evolution & CBAM · (5) Boardroom simulation

3.1 Part 1: The four forms of enterprise capital

EXPLAINED SIMPLY

Every business uses four kinds of 'wealth': machines and buildings, money, people's skills and ideas, and nature (water, soil, air, minerals). Nature is usually left off the balance sheet, but the business can't run without it.

Example: A tea company has machines (factory), money (bank balance), skills (expert tasters, brand) and nature (tea gardens, rain, healthy soil). If rains fail for three years, the other three can't save the tea crop.

Every company's balance sheet ultimately rests on four kinds of capital stock (the "Balanced Capital Matrix"):

Capital (symbol)	What it includes	Everyday picture
1. Manufactured capital (K_m)	Factories, machinery, equipment, logistics fleets, robotics, infrastructure	The "hardware" of the firm
2. Financial capital (K_f)	Cash reserves, debt facilities, equity market access, liquidity	The firm's wallet and credit card
3. Human & intellectual capital (K_h)	Talent, patents, algorithms, institutional memory, brand equity, operational code	The firm's brains and know-how
4. Natural capital (K_n)	Finite Earth assets: clean water, fertile topsoil, atmospheric carbon sinks, biodiversity, mineral reserves	Nature's "life-support system"

Brundtland definition vs. executive reality

EXPLAINED SIMPLY

The 1987 definition says: use resources today in a way that leaves enough for our children. It sounds right but doesn't tell a manager **what to do** with a budget.

Example: A CEO must choose between a cheap coal boiler and an expensive solar plant. 'Don't harm future generations' is a fine principle, but it doesn't say which boiler to buy. Managers need sharper rules.

Brundtland Commission (1987): *"Development that meets the needs of the present without compromising the ability of future generations to meet their own needs."*

Professor's verdict: an eloquent political principle, but it gives **no operational guidance** for allocating billions in capital expenditure (CapEx).

THE CORE OPERATIONAL QUESTION

Are the four forms of capital **interchangeable**? Can an executive destroy natural capital (K_n), replace it with financial or manufactured capital (K_f, K_m), and still honestly call the business "sustainable"? The answer depends on weak versus strong sustainability.

Weak sustainability: the Hartwick–Solow rule

EXPLAINED SIMPLY

The formula just says: add up all four kinds of capital each year, and the total must not go down. It doesn't matter which kind goes up or down.

Example: Your family has ₹5 lakh in savings and ₹5 lakh worth of gold. You sell gold worth ₹2 lakh and put it into savings. Total is still ₹10 lakh, so under this rule you are 'sustainable'. Weak sustainability treats a forest the same way, as if it were just gold that can be sold.

$$\frac{d}{dt}\left(K_m + K_f + K_h + K_n \right) \geq 0$$

Read it as: **"the total of all capital must not fall over time."** The "d/dt" just means "change over time".

- **Core assumption:** the mix of capital doesn't matter, because the forms are **perfect substitutes**.
- **Practical implication:** clear-cutting an old-growth forest or draining an aquifer counts as "sustainable" if the revenue builds an automated factory (K_m) or funds a university (K_h).
- **How corporate ESG applied weak sustainability over the past 20 years:** emitting CO_2 while buying cheap carbon offsets from a tree farm on another continent, and depleting municipal water tables while funding local community projects.

NUMERICAL – APPLYING THE WEAK SUSTAINABILITY RULE (ILLUSTRATIVE, USING THE CLASS FORMULA)

Question: A mining firm starts the year with K_m = 400, K_f = 200, K_h = 150, K_n = 250 (₹ crore). During the year it depletes natural capital by 120 and invests 70 in new machinery and 30 in staff training. Financial capital is unchanged. Is it "sustainable" under (a) weak and (b) strong sustainability?

1. Write the change in each capital: $\Delta K_m = +70$, $\Delta K_h = +30$, $\Delta K_f = 0$, $\Delta K_n = -120$.
2. Apply the Hartwick–Solow rule (sum of changes):

```
Δ(Total K) = 70 + 0 + 30 + (−120) = −20
```

3. **Weak sustainability test:** Is Δ(Total K) ≥ 0? −20 < 0, so **NOT sustainable**, even by the weak rule. (Had it invested 120 or more in total, it would "pass".)

4 **Strong sustainability test:** K_n fell (250 → 130). Strong sustainability asks whether the remaining natural capital is still above its **critical level (K_{crit})**. More machines can never count as a substitute, so **any loss of critical natural capital fails** regardless of the total.

Weak: total capital fell by 20, so unsustainable. Even if it had risen, strong sustainability would still reject the trade if critical natural capital was lost.

DISCUSSION: DOES VOLUNTARY CSR REPRESENT WEAK SUSTAINABILITY?

Largely, yes. Classic CSR "compensates" for harm in one place (pollution, water use) with good deeds or money elsewhere (community projects, offsets). This assumes harm to nature can be cancelled out by another type of capital, which is exactly the weak-sustainability assumption.

Strong sustainability: complements, not substitutes

EXPLAINED SIMPLY

Some things only work together, like a car and petrol, or a pen and ink. Machines and nature are like that. A machine can't make fish, timber or water; it can only process what nature provides.

Example: A sugar mill (machine) is worth nothing if the region's sugarcane fails from groundwater depletion. More machines can't fix missing cane.

Ecological economics (Herman Daly): natural capital (K_n) and manufactured capital (K_m) are **complements**, not substitutes. You cannot replace a biological system with a machine.

- **Thought experiment 1:** What is a $1 billion high-tech fishing fleet (K_m) worth if ocean fish stocks (K_n) collapse to zero? **Nearly nothing.**
- **Thought experiment 2:** What is a state-of-the-art chip fab (K_m) worth if the local river basin dries up and process water runs out? **It becomes a stranded asset.**
- **Bottom line:** advanced sawmills can never produce a single cubic foot of timber without a living forest.

Critical natural capital (K_{crit})

EXPLAINED SIMPLY

K_{crit} is the floor: the minimum amount of nature that must be kept because no money or technology can replace it. Three reasons: once lost it's gone forever, nature collapses suddenly after a tipping point, and technology must obey the laws of physics.

Example: A glacier feeding a river melts slowly for decades, then the river suddenly runs dry in summer. No amount of cash can rebuild the glacier. That is irreversibility plus a tipping point.

DEFINITION

K_{crit} is a **non-negotiable floor** of ecological assets that perform life-support functions that human technology **cannot replicate or replace**.

It rests on three physical realities:

1. **Thermodynamic irreversibility:** extinct pollinators, depleted fossil aquifers and melted glaciers cannot be "recapitalised" with cash.
2. **Non-linear tipping points:** ecosystems don't degrade in a straight line. They absorb stress until a threshold, then flip suddenly and permanently. (Think of a chair you keep leaning back on: fine, fine, fine, then you fall.)
3. **Fundamental non-substitutability:** technology works *within* the laws of thermodynamics and does not replace them.

EXECUTIVE WARNING

Physical resource limits will eventually cap balance-sheet growth, whatever the financial projections say.

Exercise: capital substitution audit (how to answer)

EXPLAINED SIMPLY

This exercise asks you to find the one natural input your business can't live without, check whether you're assuming 'technology will fix it', and test what happens if it is cut in half.

Example: Sample answer for a chip maker: (1) Critical input: ultra-pure water. (2) Assumption: we've assumed water trucks can fill any gap, which is weak sustainability. (3) Stress test: a 50% water cut stops half the production lines and margins collapse, so we must invest in water recycling now.

1. Critical input identification: what is the single most critical natural-capital input your business relies on?

Name one concrete input, such as uninterrupted power, clean water, high-purity aluminium, or unpriced carbon sinks. For a bank it might be the climate stability behind its mortgage book.

2. Underlying assumption: weak or strong?

If the plan relies on offsets or "future technology" to fix scarcity, it is **weak**. If it designs around physical limits (recycling water, securing supply, reducing dependence), it is **strong**.

3. Stress test: if that resource contracts by 50% within 36 months, does the business model survive?

Estimate the hit to output and margins. If margins collapse, the firm is exposed to a K_{crit} risk and needs resilience investment now.

3.2 Part 2: Why environment takes precedence in ESG

EXPLAINED SIMPLY

E, S and G are not equal partners. Governance (rules, boards) is created by society, and society can only exist because nature supports it. So E is the foundation of the house, S is the walls, and G is the furniture.

Example: You can replace a bad CEO (G) in a month and settle a strike (S) in a week. But if the river a city depends on dries up (E), no board meeting can fix it.

The flawed Venn diagram

EXPLAINED SIMPLY

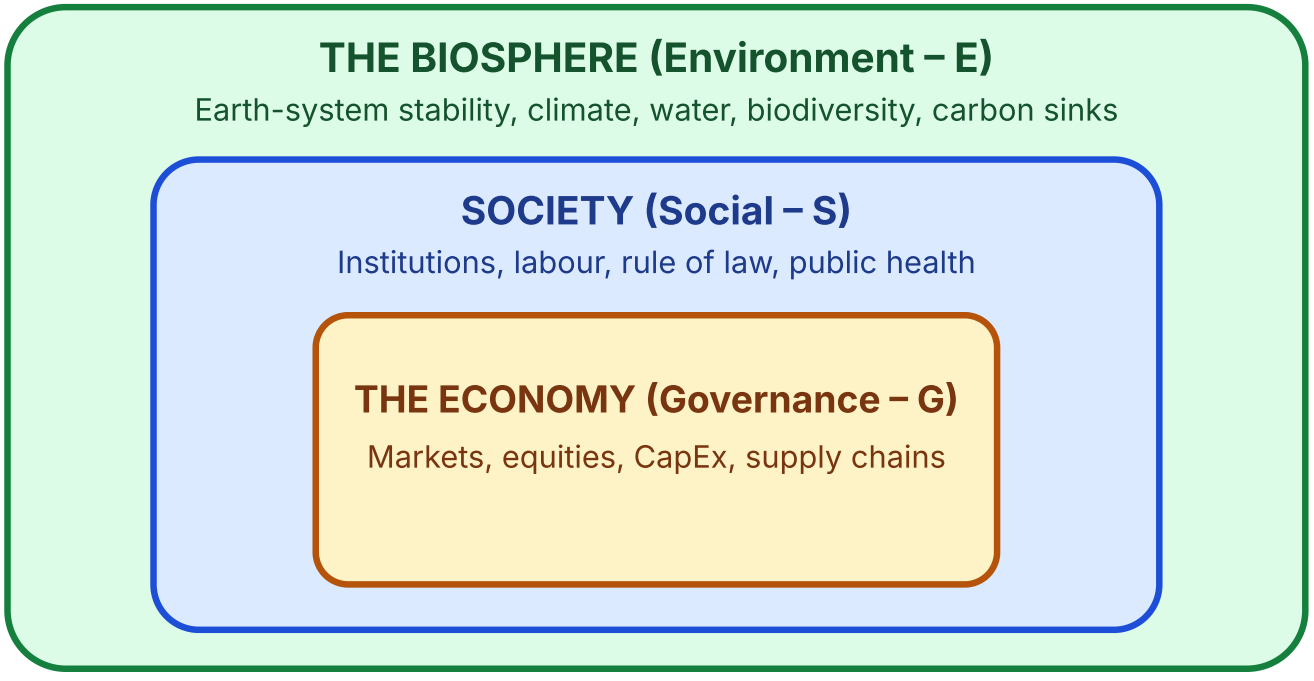
The usual picture draws E, S and G as three equal overlapping circles. That's misleading because it suggests you can trade a little nature for a little better governance as if they were equal. In reality nature is the container for everything else.

Example: It's like drawing 'fish', 'water' and 'aquarium rules' as equal circles. Without water there are no fish and no need for rules.

Textbooks often draw ESG as **three equal, overlapping circles** (E, S, G). This is misleading because:

- It implies trade-offs between E, S and G are **symmetrical**.
- It treats the physical biosphere as just another "department" alongside governance.

Scientific reality: ESG elements are **hierarchically nested**. Governance (G) is a tool created by Society (S), and Society is embedded within the Biosphere (E). **"Physics outranks contracts."**



The Nested Systems Model: the economy sits inside society, and society sits inside the biosphere.

DEPENDENCY HIERARCHY IN SIMPLE WORDS

An economy can pause (a recession) and society carries on. A society can collapse and nature regenerates. But if the **biosphere** breaks down, society collapses and market liquidity disappears. That is why E comes first.

Planetary boundaries (Rockström et al., 2009)

EXPLAINED SIMPLY

Scientists list nine 'safety limits' for the Earth, such as climate, biodiversity, forests, freshwater, nitrogen and plastics. Staying within them keeps the planet stable. We have already crossed six of the nine.

Example: Think of nine warning lights on a car dashboard. Six are already red. You can keep driving for a while, but the risk of a breakdown is now very high.

Nine Earth-system tipping points regulate the roughly 10,000-year **Holocene** climate stability that made agriculture and industrial civilisation possible.

#	Boundary	Measured by / example	#	Boundary	Measured by / example
1	Climate change	CO_2 concentration	6	Novel entities	Plastics, toxins
2	Biosphere integrity	Extinction rates	7	Stratospheric ozone depletion	Ozone layer
3	Land-system change	Deforestation	8	Ocean acidification	Ocean pH
4	Freshwater change	Soil/surface water	9	Atmospheric aerosol loading	Particles in air
5	Biogeochemical flows	Nitrogen & phosphorus	**Current status: 6 of 9 boundaries breached**: climate, biodiversity, land, water, N/P and novel entities.		

Ecological risk is un-diversifiable systemic risk

EXPLAINED SIMPLY

Money can fix governance and social problems: fire the CEO, raise wages. Physics and biology can't be bribed or negotiated with. And because a climate shock hits everyone at once, spreading your investments doesn't protect you.

Example: During a heatwave every power plant, farm and factory in the region suffers together. An investor holding shares in all of them loses everywhere at the same time.

Governance & social risks (repairable)	Ecological degradation (non-negotiable)
Fraud or governance breakdown: fire the board, write down assets, settle with regulators, restructure.	**Governed by physics and biology:** cannot be negotiated with or settled out of court.
Labour strike or backlash: renegotiate contracts, adjust wages, re-engineer culture.	**Irreversible tipping points** trigger systemic liabilities across whole markets.
Capital solution: money and institutional will can repair G and S.	**Un-diversifiable:** normal portfolio diversification cannot hedge a planetary-boundary breach.

WHY "UN-DIVERSIFIABLE"?

Investors protect themselves by spreading money across many companies, because if one fails the others make up for it. A climate shock hits **all** companies, sectors and countries together, so there is nowhere to diversify to.

Three channels through which ecological damage hits financial statements

EXPLAINED SIMPLY

Nature's damage reaches a company's accounts in three ways: insurers refuse cover (so banks refuse loans), operations stop (no water, no transport), and assets lose value early (stranded assets).

Example: Uninsurability: houses in a flood-prone area can't get insurance, so banks won't give home loans there. **Disruption:** a drought stops a textile dyeing unit. **Stranding:** a new coal plant shuts 20 years early because carbon rules make it unprofitable.

1. **Asset uninsurability:** reinsurers withdraw from high-risk regions (California, Florida). Without property insurance, banks refuse 30-year mortgages, and capital liquidity freezes.
2. **Operational disruption:** droughts forced chipmakers (TSMC) to bring in water by truck, and low Rhine River levels halted barge logistics for European heavy industry.
3. **Asset stranding:** carbon-heavy plants face accelerated write-downs as regulation penalises carbon intensity.

3.3 Part 3: Why free markets failed to solve sustainability

EXPLAINED SIMPLY

Markets didn't fix pollution for three reasons: (1) polluting was free, so cleaner firms lost out; (2) managers think in quarters and climate damage arrives in decades; (3) companies could hide the truth in glossy reports.

Example: If one bus company pays for clean CNG buses while its rival runs old diesel buses for free, the rival can charge lower fares and win passengers. Doing the right thing voluntarily was a losing strategy until rules forced everyone to pay.

Core governance question: if environmental risk is so severe, why didn't competitive markets solve it automatically, and why did decades of voluntary CSR fail to curb emissions? The answer is **three structural market failures**:

Market failure	Explanation
1. Tragedy of the commons and unpriced externalities	The atmosphere and oceans carried a **zero market price**. If Company A spends $100M to scrub emissions while Competitor B vents CO_2 for free, B has lower marginal costs. Voluntary action **penalises responsible actors and rewards free riders**.
2. Tragedy of the horizon (Mark Carney)	**Time-horizon mismatch.** Executives think in 90-day earnings and 3–5 year plans, central banks in 2–3 year inflation mandates, politicians in 4–5 year election cycles. Climate shocks compound over 10–40 years. By the time the shock hits, the executive team has retired.
3. Information asymmetry and greenwashing	When companies controlled the story through unstandardised PDF brochures, they could hide environmental liabilities behind minor green initiatives (greenwashing). The fix is the shift from **voluntary storytelling to audited discipline**: Europe's **CSRD** (Corporate Sustainability Reporting Directive) and the **ISSB** (International Sustainability Standards Board).

Legacy CSR vs. modern sustainability accounting

EXPLAINED SIMPLY

Old-style CSR was voluntary storytelling for the public, with no penalty for exaggerating. Modern sustainability reporting is audited like financial accounts, aimed at investors, and false claims can lead to lawsuits.

Example: Old: a brochure with photos of tree-planting. New: an audited report stating exactly '1.2 million tonnes CO_2, verified by auditors', with directors legally liable if it's false.

Dimension	Traditional CSR (legacy era)	Modern accounting (current era)
Audience	Consumers, NGOs, public	Investors, rating agencies, lenders
Materiality	Single materiality (firm's impact on society)	Double materiality (CSRD) / financial materiality (ISSB)
Legal standing	Voluntary PR brochures	Audited financial filings
Governing bodies	Fragmented non-profits	ISSB, CSRD
Legal liability	None (marketing risk only)	Executive liability, civil lawsuits, fraud charges

Double materiality

EXPLAINED SIMPLY

Look in both directions: **outside-in** (how climate and nature can hurt the company's money) and **inside-out** (how the company hurts nature and people). ISSB mainly asks for the first; the EU's CSRD asks for both.

> **Example:** A beverage company: outside-in, a drought could shut its plant (financial risk); inside-out, its pumping dries up farmers' wells (impact). Double materiality makes it report both.

Impact materiality (Y) →		
	Outside-in risk (current scope) high impact, low financial	**Double materiality** (CSRD mandate)
	Low priority	**Financial impact only** (ISSB scope)
	Financial materiality (X) →	

X-axis (financial materiality): how environmental change affects the *firm's* value and cash flows ("outside-in").

Y-axis (impact materiality): how the firm's operations affect the environment and society ("inside-out").

ISSB covers mainly the X-axis. **CSRD** demands both, which is **double materiality**.

WORKSHOP PROMPT ANSWERED

"If your firm only manages what is financially material today, what blind spots are you building up on the Y-axis that regulators will force onto your balance sheet within 36 months?" Examples: water drawn from stressed aquifers, supply-chain pollution, community health impacts, and Scope 3 emissions. These are "impacts" today, but carbon taxes, CBAM, lawsuits and licence withdrawals turn them into **financial** costs later. In other words, today's Y-axis items become tomorrow's X-axis items.

3.4 Part 4: The 35-year trajectory of global governance

EXPLAINED SIMPLY

Climate rules have got tougher step by step: polite promises (Rio 1992), binding targets for rich countries (Kyoto 1997), every country setting its own pledge (Paris 2015), and now taxes at the border (CBAM).

> **Example:** It's like a school moving from 'please keep the classroom clean' to 'cleaning duty for seniors only', then 'each class sets its own cleaning plan', and finally 'a fine if your class is dirty'.

Macro shift: from diplomatic goodwill and voluntary declarations → multilateral treaties → hard trade barriers, border carbon taxes and supply-chain protectionism.

Phase	Approach	Key features	Flaw / lesson
1992 Rio Earth Summit	Voluntary declarations	Created the **UNFCCC**. Two principles: **Precautionary Principle** (scientific uncertainty must not delay cost-effective prevention) and **CBDR** (Common But Differentiated Responsibilities: developed nations lead).	Aspirational, with **no enforcement** or trade sanctions.
1997 Kyoto Protocol	Top-down state quotas	**Legally binding** targets for **Annex I** (developed) nations. Early market mechanisms: **Clean Development Mechanism (CDM)** and emissions trading.	The US did not ratify and emerging powers were exempt. **Carbon leakage:** heavy industry moved from the regulated EU to unregulated markets, and global emissions rose.
2015 Paris Agreement	Bottom-up NDCs	Legally binding treaty adopted by **195 parties at COP21**. Moved from top-down quotas to **Nationally Determined Contributions (NDCs)**, with a focus on capital mobilisation.	Each country sets its own pledge.
Present: EU CBAM	Border tariffs and trade	**Carbon Border Adjustment Mechanism.** For 20 years EU firms paid for carbon under the **EU ETS** while foreign rivals exported untaxed steel and aluminium into the EU. CBAM **levels the playing field** by taxing the embedded emissions of imports at the border.	Carbon becomes a **border-adjusted tariff**.

CARBON LEAKAGE IN SIMPLE WORDS

If one country makes polluting expensive, polluting factories simply move to a country where it is free. The emissions don't disappear, they move. CBAM closes that escape route by charging at the border.

How EU CBAM works: three pillars

EXPLAINED SIMPLY

When a product enters the EU, the importer pays for the carbon emitted in making it, at the EU carbon price. If the exporter already paid a carbon tax at home, that amount is deducted. If it can't prove its emissions, the EU assumes the worst.

Example: An Indian steel exporter sends 1 ton of steel that emitted 2 tons of CO_2. At \$90/t the importer pays \$180. If India had charged \$20/t at home, the importer pays only 2 × (90 − 20) = \$140. With no data, the EU assumes a high default emission figure.

1. **Digital CBAM certificates:** importers must buy certificates matching the product's embedded emissions. Price = weekly average EU ETS auction price (e.g., \$90–\$110 per t CO_2).
2. **Home-country carbon credit deduction:** if the exporter already paid an explicit, accredited carbon price at home, that amount is **deducted** at the EU border.

3. **Penal default factors:** if the exporter can't provide audited, plant-specific emissions data, EU customs assumes the **dirtiest production process**.

NUMERICAL 1 – CBAM CASE STUDY: STEEL IMPORTS (FROM THE SLIDES)

Given: EU border carbon price = **$90 per ton CO_2e**. Two non-EU producers export steel to the EU.

Item	Producer A (coal blast furnace)	Producer B (green H_2 DRI + electric arc)
Direct emissions (t CO_2e per t steel)	2.1	0.3
Base production cost	$580 / t	$750 / t

Professor's method: CBAM fee = emissions intensity × carbon price, then landed cost = base cost + CBAM fee.

1. **CBAM fee for A:**

```
Fee(A) = 2.1 t CO2/t steel × $90/t CO2 = $189 per ton of steel
```

2. **CBAM fee for B:**

```
Fee(B) = 0.3 × $90 = $27 per ton of steel
```

3. **Total landed cost in the EU** = base cost + fee:

```
A: $580 + $189 = $769 per ton
B: $750 + $27  = $777 per ton
```

4. **Compare before and after CBAM:**

```
Pre-CBAM advantage of A  = 750 − 580 = $170 per ton
Post-CBAM advantage of A = 777 − 769 = only $8 per ton
```

The $189 fee wipes out about 95% of A's cost advantage (170 − 8 = 162 lost out of 170).

Landed cost: A = $769/t, B = $777/t. A's $170/t advantage from cheap coal shrinks to just $8/t at $90 carbon.

Extension (a): what if carbon rises to $110/t? (the slide's claim)

1.
```
Fee(A) = 2.1 × 110 = $231   →  Landed(A) = 580 + 231 = $811
Fee(B) = 0.3 × 110 = $33    →  Landed(B) = 750 + 33  = $783
```

2. Now B is cheaper by 811 − 783 = **$28 per ton**. Producer A is **priced out of Europe**, as the slide says.

Extension (b): break-even carbon price

1. Set landed costs equal, with carbon price = p:

```
580 + 2.1p = 750 + 0.3p
```

2. Solve:

```
2.1p − 0.3p = 750 − 580  →  1.8p = 170  →  p = 170 / 1.8 ≈ $94.4 per t CO2
```

Above ≈ $94.4/t CO_2, the green producer B becomes cheaper in the EU than the coal-based producer A.

Extension (c): applying the home-country carbon credit (pillar 2)

Suppose A's home country already charges an accredited carbon price of $30/t.

1. EU deducts what was already paid:

```
Net CBAM fee(A) = 2.1 × (90 − 30) = 2.1 × 60 = $126
```

2. The total carbon cost is still 2.1 × 90 = $189, but now **$63 stays with A's home treasury** (2.1 × 30) and only $126 goes to EU customs.

This is why the slide predicts that countries will set up domestic carbon pricing: to keep the tax revenue at home instead of handing it to the EU.

Strategic analysis of CBAM economics

EXPLAINED SIMPLY

Before CBAM, dirty producers were cheaper because pollution was free. CBAM adds the pollution cost back at the border, so clean producers can now compete on price.

Example: It's like a shop that sold cheap goods only because it dumped its waste on the street for free. Once the city charges for waste, its price advantage disappears.

Pre-CBAM	Post-CBAM
Producer A enjoyed an "unassailable" **$170/ton** cost advantage over B thanks to cheap local coal. Unpriced carbon let dirty producers dominate trade.	The **$189** CBAM fee wipes out A's entire cost advantage. If carbon rises to **$110/t**, A is priced out of Europe.

EXECUTIVE TAKEAWAY

CBAM turns carbon into a **border-adjusted tariff**. Governments will set up domestic carbon pricing to keep tax revenue at home rather than surrender it to EU customs.

3.5 Part 5: Boardroom simulation (how to approach it)

EXPLAINED SIMPLY

This is a role-play. Exporters must decide how to cut emissions and keep EU customers. Car makers must decide whether to switch to cleaner (but costlier) suppliers or pay the higher import costs.

> **Example:** A Tata or JSW-type exporter might invest in electric arc furnaces and buy renewable power. A BMW-type buyer might sign long-term contracts for green steel and give its old suppliers a deadline to clean up.

Group 1: Non-EU exporter	Group 2: EU automotive OEM
Profile: executive committee of a non-EU steel/aluminium conglomerate, with 30% of high-margin sales going to German and Swedish OEMs. **Task:** set a CapEx and pricing strategy to survive CBAM tariffs. **Points to raise:** (i) measure and audit plant-level emissions to avoid penal default factors; (ii) short term: renewable power PPAs and scrap-based electric arc furnaces; long term: green-hydrogen DRI CapEx; (iii) lobby for a domestic carbon price so the fee is credited at home; (iv) sign long-term green-premium contracts with OEMs; (v) redirect dirtier output to non-CBAM markets in the meantime.	**Profile:** risk and procurement committee of a European luxury-vehicle maker. **Task:** decide whether to replace non-compliant suppliers or absorb higher landed costs. **Points to raise:** (i) compare the landed-cost increase with switching costs and quality risk; (ii) a luxury brand can pass on some cost, since a ton of steel is a small share of car value; (iii) Scope 3 targets and CSRD disclosure favour low-carbon suppliers; (iv) a dual-sourcing transition: give current suppliers a deadline and co-invest in their decarbonisation; (v) secure green-steel volume early before it becomes scarce.

MODULE 4 · SLIDES (10 OCT)

Market Failures & Internalisation of Environmental Externalities

Market failure · deadweight loss · four types of externality · Pigouvian tax/subsidy · command-and-control · asymmetric information · Bhopal, Zara, Starbucks

4.1 What is market failure?

EXPLAINED SIMPLY

A market 'fails' when buying and selling freely gives a result that's bad for society as a whole: too much of something harmful, too little of something useful, or prices that are too high.

Example: Firecrackers on Diwali: buyers pay the shop price, but the smog that harms everyone isn't in that price. So people buy more than is good for society. The market has failed.

DEFINITION

A situation where the free allocation of goods and services by a market is **not efficient**, causing a net loss of social welfare. In symbols: **MSC ≠ MSB** (marginal social cost ≠ marginal social benefit) at the market quantity.

Primary causes:

- **Externalities:** costs or benefits imposed on third parties (pollution, vaccination).
- **Public goods:** non-excludable and non-rivalrous, which leads to the **free-rider problem**.
- **Asymmetric information:** adverse selection and moral hazard distort transactions.
- **Market power:** monopolies and oligopolies restrict output to raise price (**P > MC**).

KEY TERMS IN PLAIN WORDS

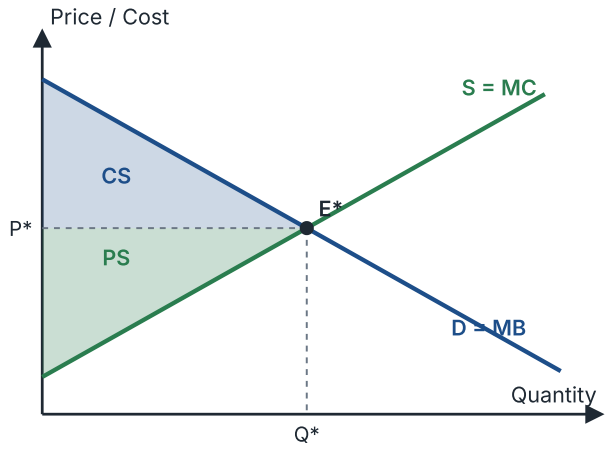
MB (marginal benefit) = how much people value one more unit, which is the **demand curve**. **MC (marginal cost)** = the cost of producing one more unit, which is the **supply curve**. **Consumer surplus (CS)** = what buyers would have paid minus what they actually pay. **Producer surplus (PS)** = price received minus cost. **Deadweight loss (DWL)** = welfare that is simply lost, a "lost slice of the cake" that nobody gets.

4.2 Efficiency vs. deadweight loss

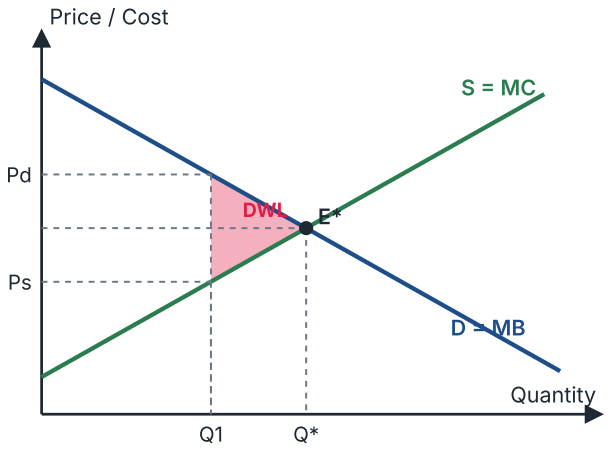
EXPLAINED SIMPLY

The 'right' amount of anything is where the value of one more unit equals the cost of making it. Producing less or more than that wastes value. The waste is called **deadweight loss**.

> **Example:** If people would happily pay ₹50 for a cup of tea that costs ₹20 to make, but the only tea stall limits sales, those unmade cups are lost happiness. Nobody gains it; it simply disappears.



Competitive equilibrium: total surplus is maximised



Under-production (Q1 < Q*): MB > MC, deadweight loss

- **Competitive equilibrium (efficient):** where **MB = MC**, at P*, Q*. Total surplus (CS + PS) is **maximised**.
- **Underproduction / monopoly (inefficient):** output Q_1 < Q* means **MB > MC** for the units between Q_1 and Q*. People value them more than they cost, yet they aren't produced. The triangle between the curves is the **DWL**. Buyers pay P_d and sellers receive P_s.

4.3 Externalities

EXPLAINED SIMPLY

An externality is a side-effect on someone who wasn't part of the deal. If it's harmful it is negative; if it's helpful it is positive. Because the price ignores it, the market makes too much of the bad things and too little of the good things.

> **Example: Negative:** a factory's smoke makes nearby residents ill. **Positive:** your neighbour's beautiful garden makes your street nicer and your house worth more, but you don't pay for it.

DEFINITION

An **externality** is a cost or benefit from an activity that affects an otherwise **uninvolved third party** without being reflected in market prices. In short: **social value/cost ≠ private value/cost**.

	Negative externality	Positive externality
Definition	Imposes an **uncompensated spillover cost** on others	Generates an **unpaid benefit** for others
Market outcome	Market **overproduces** relative to the social optimum	Market **underproduces** relative to the social optimum

THE FOUR FORMULAS TO MEMORISE

```
Negative production:   MSC = MPC + MEC     (e.g., factory pollution)
Negative consumption:  MSB = MPB − MEC     (e.g., second-hand smoke)
Positive production:   MSC = MPC − MEB     (e.g., R&D spillovers)
Positive consumption:  MSB = MPB + MEB     (e.g., vaccination, education)
```

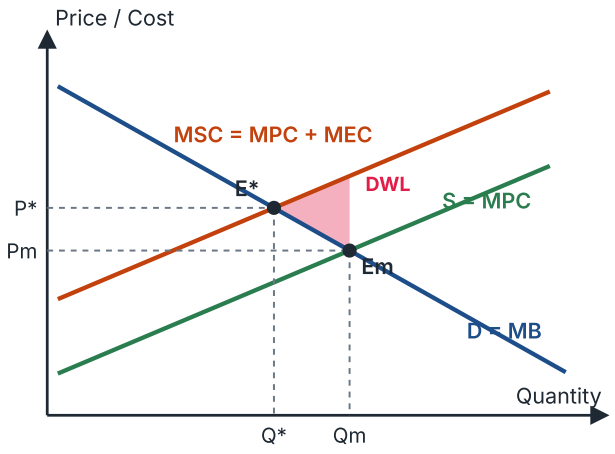
MPC/MPB = marginal **private** cost/benefit. MSC/MSB = marginal **social** cost/benefit. MEC = marginal **external cost**. MEB = marginal **external benefit**.

(1) Negative production externality

EXPLAINED SIMPLY

The factory pays for raw materials and labour but not for the damage its pollution does. So its costs look lower than they really are, and it produces too much.

Example: A leather tannery dumps chemicals in a river. Its cost per hide is ₹100, but the damage to farmers downstream is another ₹30. Society's true cost is ₹130, yet the tannery prices as if it were ₹100.



Negative production externality: over-production

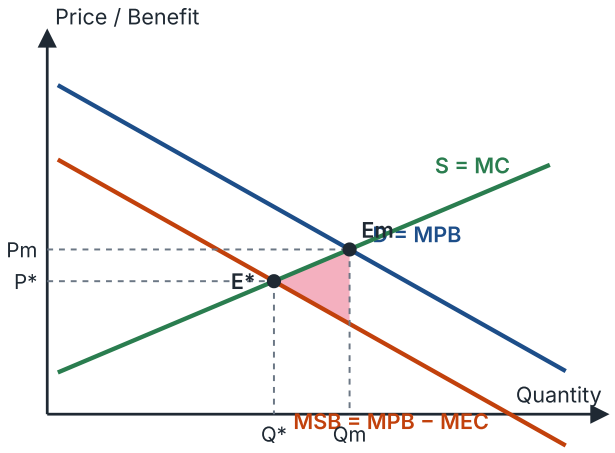
1. **Private market:** firms consider only their own cost, MPC. Equilibrium E_m is where D = MPC, giving Q_m, P_m.
2. **Social cost is higher:** pollution adds MEC, so MSC = MPC + MEC. The efficient point E* is where D = MSC, so **$Q^* < Q_m$** and $P^* > P_m$.
3. **Overproduction:** the units from Q* to Q_m cost society more than they are worth. The shaded triangle is the **DWL**.
4. **Remedy:** a **Pigouvian tax equal to MEC**.

(2) Negative consumption externality

EXPLAINED SIMPLY

Here the harm comes from *using* the product. The buyer enjoys it, but others suffer, so people consume more than is good for society.

Example: A smoker enjoys a cigarette, but people nearby breathe the smoke. Someone playing loud music late at night is another example. That's why we have public smoking bans and noise rules.



Negative consumption externality: over-consumption

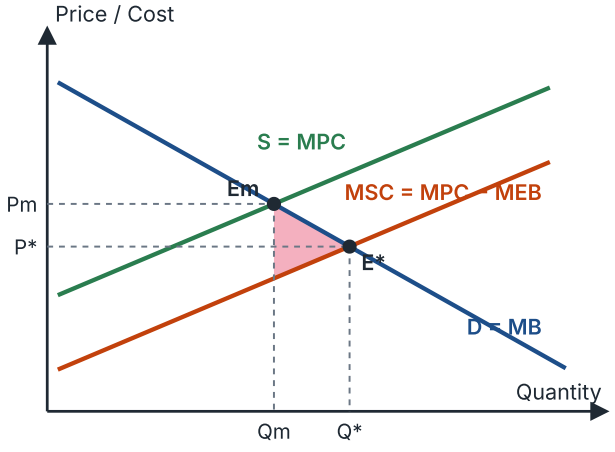
1. **Private market:** consumers consider only their own benefit, MPB. E_m is where MPB = MC.
2. **Social benefit is lower:** consumption harms others (second-hand smoke), so MSB = MPB − MEC. E* is where MSB = MC, so **$Q^* < Q_m$**.
3. **Overconsumption:** the shaded triangle is the DWL.
4. **Remedy:** a tax on consumption, or regulation.

(3) Positive production externality

EXPLAINED SIMPLY

Some production creates benefits for others that the producer isn't paid for. Because the producer gets no reward for these spill-overs, it produces too little.

Example: A company does research on drought-resistant seeds. Other firms and farmers learn from it for free. Society gains more than the company earns, so a subsidy for R&D makes sense.



Positive production externality: under-production

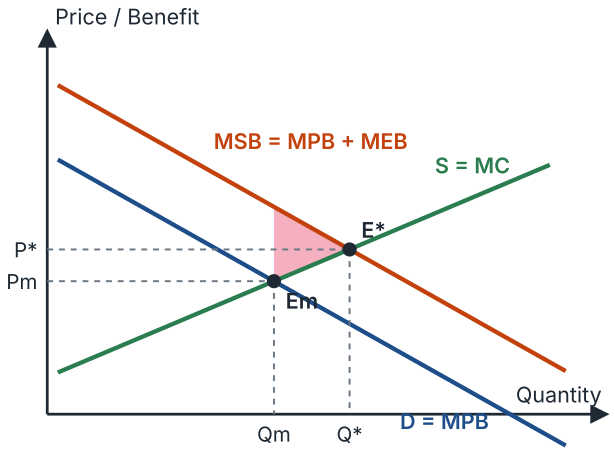
1. **Private market:** firms consider only MPC. E_m is where D = MPC.
2. **Social cost is lower:** production creates outside benefits (R&D spillovers), so MSC = MPC − MEB. E* is where D = MSC, so **$Q^* > Q_m$**.
3. **Underproduction:** the shaded triangle is the DWL (forgone surplus).
4. **Remedy:** a **per-unit subsidy equal to MEB**.

(4) Positive consumption externality

EXPLAINED SIMPLY

Some things help others when *you* use them. People only weigh their own benefit, so they use too little of these things.

> **Example:** When you get vaccinated you protect yourself *and* the people around you. When a child is educated, society gets a better citizen. That's why governments give free vaccines and free schooling.



Positive consumption externality: under-consumption

1. **Private market:** consumers consider only MPB. E_m is where MPB = MC.
2. **Social benefit is higher:** vaccination and education benefit others, so MSB = MPB + MEB. E* is where MSB = MC, so **$Q^* > Q_m$**.
3. **Underconsumption:** the shaded triangle is the DWL (forgone surplus).
4. **Remedy:** a subsidy on consumption, or public provision.

ONE TRICK TO REMEMBER ALL FOUR

Negative → too much → tax it. Positive → too little → subsidise it. The DWL triangle always sits between the private and the social curve, between Q_m and Q*.

NUMERICAL 2 – PIGOUVIAN TAX ON A NEGATIVE PRODUCTION EXTERNALITY (PRACTICE PROBLEM USING THE CLASS METHOD)

Question: Demand (MB): P = 100 − Q. Private supply (MPC): P = 20 + Q. Each unit causes pollution damage (MEC) of 20. Find (a) market output and price, (b) socially optimal output and price, (c) the deadweight loss, and (d) the correct Pigouvian tax.

1. **Market equilibrium** (D = MPC):

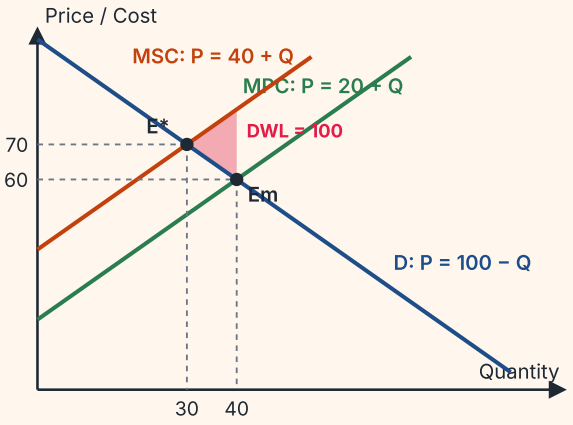
```
100 − Q = 20 + Q
2Q = 80  →  Qm = 40
Pm = 100 − 40 = 60
```

2. **Build MSC** (MSC = MPC + MEC):

```
MSC = (20 + Q) + 20 = 40 + Q
```

3. **Social optimum** (D = MSC):

```
100 − Q = 40 + Q
2Q = 60  →  Q* = 30
P* = 100 − 30 = 70
```



Worked example: Pigouvian tax t = 20

4. **DWL** = area of the triangle between MSC and D from Q* to Q_m. Its height at Q_m is MSC(40) − D(40) = 80 − 60 = 20 (= MEC). Its base is Q_m − Q* = 10.

```
DWL = ½ × base × height = ½ × 10 × 20 = 100
```

5. **Pigouvian tax** t* = MEC(Q*) = **20 per unit**. The new private supply becomes MPC + t = 20 + Q + 20 = 40 + Q = MSC, so the market moves to Q = 30. Overproduction is eliminated and DWL = 0.

```
Tax revenue = t × Q* = 20 × 30 = 600
```

Qm = 40 at P = 60; Q* = 30 at P = 70; DWL = 100; Pigouvian tax t* = MEC = 20 per unit, raising 600 in revenue.

NUMERICAL 3 – PIGOUVIAN SUBSIDY ON A POSITIVE CONSUMPTION EXTERNALITY (PRACTICE PROBLEM)

Question: Private demand (MPB): P = 100 − Q. Supply (MC): P = 20 + Q. Each unit consumed (e.g., a vaccine dose) gives others a benefit (MEB) of 10. Find Q_m, Q*, the DWL and the right subsidy.

1. Market (MPB = MC): 100 − Q = 20 + Q, so **Q_m = 40**, P_m = 60.
2. Build MSB = MPB + MEB = 110 − Q.
3. Optimum (MSB = MC): 110 − Q = 20 + Q, so 2Q = 90 and **Q* = 45**.
4. DWL (forgone surplus) = ½ × (Q* − Q_m) × MEB = ½ × 5 × 10 = **25**.
5. Subsidy s* = MEB(Q*) = **10 per unit**. Effective demand becomes MPB + 10 = MSB, so consumption rises to 45. Subsidy cost = 10 × 45 = 450.

Market under-consumes (40 vs 45); DWL = 25; a subsidy of 10 per unit fixes it.

4.4 Correcting externalities: policy solutions

EXPLAINED SIMPLY

Two ways to fix it: (1) change the price, by taxing harmful things or subsidising helpful ones, or (2) set rules directly, through limits, bans or mandates.

Example: Price route: a 'green cess' per tonne of coal, or a subsidy on rooftop solar. **Rules route:** a ban on single-use plastic, or a legal limit on how much smoke a chimney may emit.

1. Price-based	2. Command-and-control
Pigouvian tax: per-unit tax equal to marginal external damage, **t* = MEC(Q*)**. It shifts private supply up (MPC + t = MSC) and eliminates overproduction. **Pigouvian subsidy:** per-unit subsidy equal to marginal external benefit, **s* = MEB(Q*)**. It lowers cost to expand output to Q*.	**Production controls and quotas:** an explicit output cap or emission ceiling (Q ≤ Q*). **Consumption controls and mandates:** restrict consumption directly (smoking bans in public places, zoning), or mandate it for positive externalities (compulsory primary schooling, vaccine mandates).

4.5 Limits and challenges of interventions: "government failure"

EXPLAINED SIMPLY

Government fixes can go wrong too. It's hard to put a price on clean air, taxes can hurt the poor most, the tax may be set too high or too low, uniform rules ignore that some firms can clean up more cheaply than others, and checking everyone is expensive.

Example: A petrol tax hurts a delivery boy (for whom fuel is a big share of income) more than a rich car owner. A rule that every factory must buy the same filter gives nobody a reason to invent a better one.

Core dilemma: correcting market failures requires accurate valuation of unpriced damage. In practice, regulatory friction, information gaps and political incentives can cause large welfare losses.

Limits of market-based solutions (taxes and subsidies)	Limits of command-and-control (mandates and quotas)
• **Valuing non-market goods:** putting a money figure on clean air, biodiversity or health is subjective. • **Secondary deadweight loss:** if t ≠ t*, output is over- or under-restricted, creating a new inefficiency. • **Elasticity uncertainty:** if price elasticity of demand is uncertain, a fixed tax can't guarantee a specific quantity reduction. • **Regressive incidence:** taxes on essentials (fuel) take a larger share of poorer households' income. • **Distortionary funding:** subsidies need tax revenue raised through distortionary taxes elsewhere. • **Political obstacles:** public resistance to taxes; lobbying and rent-seeking for subsidies.	• **Different abatement costs:** firms have different marginal abatement costs (MAC). A uniform cap forces high-cost firms to abate too much and low-cost firms too little. • **Dynamic inefficiency:** technology standards (e.g., requiring a specific filter) remove the incentive to innovate beyond the minimum. • **Heavy information burden:** regulators need detailed technical data on every plant. • **High surveillance costs:** checking continuous compliance needs costly monitoring networks.

NUMERICAL 4 – WHY UNIFORM CAPS ARE INEFFICIENT (ILLUSTRATING "HETEROGENEOUS ABATEMENT COSTS")

Question: Two firms each emit 100 t. The regulator wants a total cut of 100 t. Firm L can abate at $10/t and Firm H at $50/t (constant costs). Compare (a) a uniform cap (each cuts 50 t) with (b) a price-based approach where each firm faces a tax of $30/t.

1. **Uniform cap:** cost = 50 × $10 + 50 × $50 = $500 + $2,500 = **$3,000**.
2. **Tax of $30/t:** L abates whenever its cost ($10) < tax ($30), so it cuts all 100 t. H's cost ($50) > tax ($30), so it pays the tax and doesn't abate. Total cut = 100 t. Abatement cost = 100 × $10 = **$1,000**.
3. Saving to society = 3,000 − 1,000 = **$2,000** for the same environmental result.

The uniform cap costs 3× as much because it ignores differences in MAC. Price-based tools let the cheapest abater do the work.

4.6 Asymmetric information

EXPLAINED SIMPLY

One side knows more than the other. **Adverse selection** is the problem *before* a deal (you can't spot the bad apples). **Moral hazard** is the problem *after* a deal (people behave carelessly once they're protected).

Example: Before: in a second-hand bike market buyers can't tell good bikes from bad, so good bikes stop being sold. **After:** once your car is fully insured you may park it carelessly.

DEFINITION

One party to a transaction has greater or more accurate material knowledge than the other. This distorts incentives and prevents mutually beneficial trades.

Adverse selection (hidden information, before the deal)	Moral hazard (hidden action, after the deal)
• **Akerlof's "Market for Lemons" (1970):** buyers will pay only for average expected quality, so sellers of high-quality goods exit and the market unravels. • **Insurance death spirals:** flat premiums make low-risk people opt out, raising average claims and premiums until only high-risk buyers remain. • **Welfare impact:** missing markets or severe under-trading of high-quality goods. • **Remedies: signalling (Spence 1973)**, where the informed party invests in a costly, verifiable signal (degrees, warranties, third-party certifications); and **government mandates** such as compulsory auto or basic health insurance to stop adverse pooling.	• **Insurance-induced neglect:** policyholders take less care, or overuse healthcare when its marginal private cost is near zero. • **Principal–agent friction:** managers (agents) pursue perks or shirk because shareholders (principals) can't monitor them perfectly. • **Remedies: cost-sharing** (deductibles, copayments, coinsurance); **incentive contracts** (stock options, bonuses, piece rates tied to outcomes); **monitoring and auditing**.

IN SIMPLE WORDS

Adverse selection: you can't tell a good used car from a bad one, so you offer an average price, and the owners of good cars refuse to sell. **Moral hazard:** once your phone is insured, you're a little less careful with it.

Information asymmetry in corporate sustainability: the "sustainable information wedge"

EXPLAINED SIMPLY

Companies know their true pollution; outsiders don't. So dirty companies can pretend to be green (green lemons), and managers can make big promises for 2050 that someone else will have to keep (hidden action).

Example: Two FMCG firms both print 'eco-friendly' on their packs. Only one actually changed its plastic. Shoppers can't tell, so both get the green image. A CEO who promises 'net zero by 2050' gets praise today but will have retired long before 2050.

Firms hold private information about their real carbon footprints, supply-chain compliance and climate transition risks that investors, consumers and regulators can't easily verify.

- **Adverse selection (ex-ante): "Green lemons."** A **greenwashing pooling equilibrium**: low-abatement firms mimic green leaders through superficial PR and selective disclosure, without spending real CapEx. Outsiders can't tell them apart.
- **Moral hazard (ex-post): hidden action. Long-horizon pledges:** managers book immediate valuation gains from "Net-Zero 2050" pledges and push the actual work onto future executives.

4.7 When governance fails: other ways to internalise externalities

EXPLAINED SIMPLY

If the government can't or won't fix an externality, other routes exist: the affected parties bargain directly (Coase), communities manage the resource themselves (Ostrom), NGOs and certifications expose bad firms, or industries agree common standards among themselves.

Example: A factory and nearby fishermen agree that the factory pays the fishermen's co-operative to keep the lake clean (Coase). Consumers boycott a brand after an NGO exposes child labour (civil pressure).

1. Decentralised and community governance	2. Market, legal and civil mechanisms
Coasean bargaining (property rights): if property rights are well defined and transaction costs are low, private parties can bargain directly to an efficient outcome without the state. **Polycentric and CPR systems (Ostrom):** communities govern shared resources with local rules (see Module 7).	**Private certification and boycotts:** NGO scrutiny, third-party audits and consumer or investor boycotts impose reputational and financial penalties. **Voluntary cross-firm compacts:** industry-wide standards and self-policing agreements to prevent a race to the bottom.

4.8 Case 1: Bhopal (1984), acute shock and unpriced damage

EXPLAINED SIMPLY

Union Carbide saved money by cutting safety. That made its own costs look low, but the true cost to society was enormous: thousands of deaths and harm that continues across generations. The company never paid the full bill.

Example: It's like a bus owner who saves money by never servicing the brakes. His profits look good until the crash, and then the passengers and their families pay the real price.

THE FACTS

In December 1984, over **40 tonnes of methyl isocyanate (MIC)** leaked from **Union Carbide's** pesticide plant in Bhopal, killing an estimated **8,000+ people in the first week** and exposing over **500,000 residents**.

Private cost-cutting decisions	Immediate economic impact
The plant ran on reduced safety budgets, with sub-standard refrigeration and malfunctioning scrubbers. Its **private marginal cost (MPC) was artificially lowered** by skipping critical safety maintenance.	A **massive divergence between MPC and MSC**: total destruction of human capital and livelihoods. **Severe information asymmetry:** officials initially withheld toxicology data and downplayed MIC's lethal reactivity.

- **Intergenerational externality:** the damage didn't end with the 1984 victims. Ongoing exposure shifted the social damage (MEC_t) onto generations born decades after the plant closed.
- **Institutional failure and corporate shielding:** when property rights, tort law and national regulation fail, a multinational can avoid full remediation through corporate restructuring and jurisdictional shielding.

LESSON

Saving a little on safety (low MPC) created enormous social cost (MSC). Because the company never paid the full cost, the externality stayed **unpriced**. This is the textbook case of complete governance and tort failure.

4.9 Case 2: Zara / Inditex, the sustainability dilemma of fast fashion

EXPLAINED SIMPLY

Zara is famous for selling trendy clothes fast and cheap. Being truly green would raise its costs, but a rival (Shein) is even faster and cheaper. Zara is squeezed between pressure to clean up and pressure to cut prices.

Example: It's like a street-food stall asked by health inspectors to switch to costlier, cleaner oil while the stall next door keeps using cheap oil and sells at half the price.

THE CORE QUESTION

How should Zara adapt when caught between two opposing forces, and which stakeholders matter most for its long-term competitive advantage?

"Slow down and clean up"	"Speed up and cut prices"
Driven by eco-conscious consumers, climate activists and ESG asset managers. Demands decarbonisation, toxic-chemical removal, higher wages and circularity. **Impact:** raises marginal cost (MPC → MSC) and threatens hyper-turnover.	Driven by ultra-fast digital entrants, mainly **Shein**. **Impact:** punishes higher costs and erodes price-sensitive market share.

Background: scale and evolution

EXPLAINED SIMPLY

Inditex grew from one Spanish shop into the world's biggest fashion group, and Zara earns most of its money. In 2022 it launched a small 'sustainable' collection, a sign it was starting to respond to pressure.

Example: Think of a small family shop growing into a national chain. It eventually has to answer to big investors and regulators, not just to local customers.

- Founded in **1963 by Amancio Ortega** in Spain. First Zara store in **1975 (La Coruña)**; international from **1988 (Porto)**.
- Grew from **622 stores and €1.2 bn revenue** (two decades earlier) to **€27.7 bn in 2021**.
- 8 banners: Zara, Pull & Bear, Massimo Dutti, Bershka, Stradivarius, Oysho, Zara Home, Uterqüe. **Zara generates 66% of net sales and 70% of earnings.**
- **Sustainability pivot (June 2022):** first limited-edition "sustainable" capsule, made with polyester yarn from **captured carbon emissions**, a notable shift for the pioneer of fast fashion.

Source of advantage: design and manufacturing agility

EXPLAINED SIMPLY

Zara's superpower is speed. Designers sit next to buyers, store sales data comes back daily, and most production is close to Spain, so a new design reaches stores in about 15 days instead of months.

Example: If a celebrity wears a new style on Monday, Zara can have something similar in stores within about three weeks, while a typical brand would need 3–4 months.

- **700+ designers** at HQ in Arteixo, producing about **65,000 designs a year**. Design, commercial and procurement teams sit together.
- **15 business days** from design to rack, versus **16 weeks** for rivals' seasonal lines and **6–9 months** for basics.
- Feedback loops: hard sales data from **RFID** garment chips plus soft input from store staff.
- **59% near-shoring** (Spain, Portugal, Morocco, Turkey, many company-owned). The global network has **824 suppliers, 7,210 factories, 47 countries**. Distant factories make basic, undyed fabric. Plants

run a **4.5-day week** to keep surge capacity.

Logistics, the "Zara gap", and prime real estate

EXPLAINED SIMPLY

Zara restocks twice a week in small batches, so customers feel 'buy it now or it's gone'. As a result it rarely needs big discounts (the 'Zara gap'), and it spends on prime store locations instead of advertising.

Example: Limited 'drops' of sneakers sell out instantly at full price. Zara uses the same scarcity trick on everyday clothing.

- Restocks reach stores **twice weekly within 48 hours** via 10 Spanish logistics hubs. Selective **air freight (15–36% of inbound)** costs **12–16× ocean shipping** but protects responsiveness. Small batches create urgency ("buy it now or never").
- **The "Zara Gap" (Hausman, Stanford):** Zara discounts only **≈15% of items**, versus **50–70%** markdowns at typical US department stores.
- **Flagship stores as billboards:** heavy investment in prime real estate. Ad spending is about **one-tenth** of the industry benchmark of 3–4% of revenue (≈0.3–0.4%).

Competitive benchmark

EXPLAINED SIMPLY

Compared with rivals: Gap is slow and discounts heavily, H&M is bigger but slower, and Shein is far faster (thousands of new items daily) and sells only online through social media.

Example: If Zara is a fast restaurant kitchen, Shein is a cloud kitchen launching a new dish every few minutes and advertising it through Instagram influencers.

Dimension	Gap Inc.	H&M	Shein
Sourcing	>90% outsourced to Asia and Latin America	800 commercial suppliers (60% in Asia)	Network of small workshops in Guangdong, China
Lead time	10 weeks to 6+ months; heavy markdowns ("Gap Cash")	Slower restocks; larger assortments than Zara	Ultra-fast: **6,000 new items/day** (Zara ≈ 600)
Retail format	Brick-and-mortar retrenchment	Global stores + expanding online	Pure-play online, powered by TikTok and influencers
Valuation / status	Store closures; shrinking margins	Stagnating profits (2017 slump); pushed RFID	Valued at **$100 bn** in 2022 (vs Inditex **$77 bn**)

Emerging challengers: resale and rental platforms such as ThredUp, TheRealReal, Rent the Runway and Depop.

The externality: ecological damage in numbers (Morningstar)

EXPLAINED SIMPLY

Fast fashion pushes people to buy often and throw clothes away quickly. The environmental cost (carbon, dye pollution, microplastics, landfill) is paid by society and cities, not by the brand.

Example: A ₹400 T-shirt worn 7 times and thrown away still leaves dye in a river and plastic fibres in the sea. The city then pays to collect and dump the waste.

Fast fashion sells about the same total volume as traditional retail, but rapid turnover drives impulse over-consumption and early disposal.

Upstream production costs	Downstream municipal waste
Fashion ≈ **10% of global carbon emissions**. **20% of global industrial wastewater** comes from textile dyeing and treatment. Synthetics release **190,000 tonnes of microplastics** into oceans every year.	**85%** of clothing bought by US consumers ends up in landfills or incinerators. Rapid wardrobe turnover shifts waste-collection costs onto **municipal budgets**. Garments discarded after only **7–10 wears** create an unpriced, cross-border waste stream.

Rising pressures: activists and capital markets

EXPLAINED SIMPLY

Pressure on fashion brands has moved from protesters outside shops to big investors in boardrooms, who can vote out directors who ignore climate.

Example: When BlackRock, one of the world's largest investors, says it will vote against board members who ignore climate, directors pay attention far more quickly than they would to a street protest.

- **Activists and civil society:** *The True Cost* documentary (2015) exposed supply-chain problems. Sunrise Movement and Extinction Rebellion mobilised directly. In 2021 activists stormed a **Louis Vuitton** catwalk.
- **Institutional capital: 33%** of environmental and social shareholder resolutions passed in 2022 (up from **22%** five years earlier). **BlackRock** warned it would vote against directors who fail on climate oversight. ESG compliance moved from public relations to fiduciary necessity.

Zara's commitments vs. business realities

EXPLAINED SIMPLY

Zara promised greener materials (recycled polyester, sustainable cotton). Critics say this is like switching to diet cola while still drinking ten bottles a day: the real problem is the sheer volume of short-lived clothes.

Example: Making 1 billion garments from recycled plastic is still 1 billion garments that will be thrown away quickly. The business model, not just the fabric, is the externality.

Inditex's 2019 blueprint: Net Zero by **2040**; 100% sustainable cotton; zero landfill waste in offices and logistics; zero single-use plastics by 2023; 100% recycled polyester and 25% water reduction by 2025.

- **Restructuring costs:** overhauling supplier relationships needs heavy, sustained CapEx, which raises unit costs.
- **The greenwashing critique:** swapping virgin for recycled polyester cleans up raw materials but ignores the **business engine**. It fails to address the core externality: **stimulating continuous high-volume sales of short-lived clothing**.

Strategic tensions

EXPLAINED SIMPLY

Young shoppers say they care about the planet but still buy cheap, fast fashion in bulk (the Gen Z paradox). And if Zara spends more on green fabric, Shein can undercut it even further.

Example: A student shares climate posts on Instagram, then orders a 'haul' of ten ₹300 outfits. What people say and what they actually buy are different.

- **The Gen Z paradox:** Gen Z ranks climate change as a top concern and drives traffic to resale apps, yet it is the main engine of ultra-fast-fashion "haul culture". **Stated preferences conflict with revealed purchasing behaviour.**
- **Cost disadvantage vs. Shein:** certified sustainable fabrics raise Zara's costs and open a window for Shein. A Zara dress at **$79** competes with a Shein substitute at **$15**.

QUICK CALCULATIONS FROM THE ZARA CASE

1. **Price gap:** 79 ÷ 15 ≈ **5.3×**. Zara's dress costs about 427% more (64 ÷ 15) than Shein's.
2. **Speed gap:** Shein's 6,000 new items a day vs Zara's ≈ 600 is **10×**.
3. **Revenue growth:** €1.2 bn → €27.7 bn ≈ **23×**.
4. **Markdown advantage:** Zara marks down 15% of items vs 50–70% for rivals, so it sells 85% of items at full price against only 30–50%.

THE ECONOMICS IN ONE LINE

This is a **prisoner's dilemma**. If Zara alone pays to internalise its externalities (MPC rises toward MSC) while Shein doesn't, Zara loses price-sensitive customers. Voluntary internalisation is trapped by ultra-fast discounters and high volume.

4.10 Case 3: Starbucks, "performance through humanity"

EXPLAINED SIMPLY

Starbucks realised that without healthy farms and farmers it would have no good coffee to sell. So paying for sustainability wasn't charity; it was protecting its own supply.

Example: A mango-juice company that helps orchard farmers fight pests and drought is really protecting next year's mangoes.

The economic rationale	The core strategic pillars
Coffee supply is vulnerable to climate change, soil depletion and smallholder poverty. Externalities are treated not as philanthropy but as **existential supply-chain threats**. Securing long-term supply requires **internalising social and ecological costs directly into procurement**.	**Ethical sourcing:** farmer resilience, access to credit, sound ecosystem management. **Green retail:** greening energy, water and building standards. **Human capital and community:** youth pathways, employee healthcare, shared economic value.

Solving information asymmetry: C.A.F.E. Practices

EXPLAINED SIMPLY

Buyers can't see how coffee was grown. Starbucks created C.A.F.E. Practices, a checklist audited by outsiders covering quality, fair pay, worker safety and environmental care, so customers can trust the claim.

Example: It's like an 'ISI mark' or 'organic certified' label. A trusted third party checks the claim so you don't have to take the seller's word.

C.A.F.E. (Coffee and Farmer Equity) Practices rest on four standards:

1. **Product quality:** the baseline for premium pricing.
2. **Economic transparency:** mandatory proof that payments reach smallholders and farm workers.
3. **Social responsibility:** safe conditions, fair wages, no child labour.
4. **Environmental leadership:** soil conservation, biodiversity, water protection, chemical controls.

Scaling up: 1998, only 30–40 million lbs certified with Conservation International (CI). 2004 launch: **43 million lbs** bought under C.A.F.E. (**14.5%** of purchases). 2014: **96%** of all coffee ethically sourced (over 400 million lbs), 95.5% via C.A.F.E. and 8.6% Fairtrade.

Upstream capital: open-source agronomy and credit

EXPLAINED SIMPLY

Starbucks helps farmers directly with training centres, better plant varieties and cheap loans, because small farmers often can't get fair credit from banks.

Example: A dairy co-operative like Amul gives farmers vet services and cattle-feed loans so they produce more and better milk, which benefits both the farmers and the brand.

- **Farmer Support Centers** in Costa Rica, Rwanda, Tanzania, Ethiopia, Colombia and China give hands-on training to raise yields, cut costs and reduce chemical runoff.
- **Hacienda Alsacia** (Costa Rica, bought 2013): an R&D farm developing climate- and rust-resistant varieties.
- **Correcting capital-market failure:** smallholders face credit rationing and exploitative lending. Loans were expanded to **$16.3 million in 2014 (+27% over 2013)**, targeting **$20 million by 2015**. Micro-loans fund wet-mill upgrades and climate adaptation.

Greening downstream operations: triumphs and frictions

EXPLAINED SIMPLY

Inside its own stores Starbucks could easily go green (green buildings, renewable power, less water). Changing customers' habits was much harder: few people bring their own cup.

Example: A college canteen can switch to solar easily, but getting students to bring their own steel tumbler is far harder.

Successes	Friction from consumer behaviour
Over **500 LEED stores in 19 countries** by 2014, more than any other retailer. Renewable energy equal to **59.3%** of global company-operated power (74% in US/Canada). **23.1% drop** in store water use vs the 2008 baseline.	**Tumbler bottleneck:** goal of 5% of drinks in reusable tumblers; achieved only **1.8%** in 2014 despite $1 reusable cups. **Infrastructure limits:** front-of-store recycling reached only **47%** of US/Canada stores (landlord resistance, gaps in municipal recycling). **Lesson:** internal operations can be controlled directly; shifting customer habits is much harder.

QUICK CALCULATIONS FROM THE STARBUCKS CASE

1. **Total coffee purchased in 2004:** 43 million lbs was 14.5% of purchases, so total = 43 ÷ 0.145 ≈ **296.6 million lbs**.
2. **2013 loan level:** $16.3 M is 27% above 2013, so 2013 = 16.3 ÷ 1.27 ≈ **$12.8 million**. Growth needed to hit the 2015 target: (20 − 16.3) ÷ 16.3 ≈ **22.7%**.
3. **Tumbler goal achievement:** 1.8 ÷ 5 = **36%** of the target. A clear sign that consumer behaviour is the bottleneck.

4.11 Synthesis: comparing the three cases

EXPLAINED SIMPLY

Bhopal: no one forced the firm to pay, and disaster followed. Zara: wants to go green, but competition punishes it. Starbucks: going green directly protects its product, and customers pay a premium for it, so it succeeds.

> **Example:** Sustainability sticks when it makes business sense (Starbucks). It struggles when it only adds cost in a price war (Zara). It fails badly when there is no accountability at all (Bhopal).

Case	What happened to the externality	Why
Bhopal	Complete governance and tort failure → **unpriced, catastrophic, intergenerational damage**	Safety costs cut, information hidden, legal shielding
Zara	**Prisoner's dilemma** → voluntary internalisation trapped by ultra-fast discounters (Shein) and high volume	Green costs raise prices in a price-sensitive market
Starbucks	**Successful self-regulation** → sustainability built directly into long-term strategy	See the three reasons below

WHY STARBUCKS SUCCEEDED WHERE OTHERS STRUGGLED

1. **Direct strategic complementarity:** high-quality coffee cannot exist without healthy ecosystems and thriving farmers, so sustainability *is* the business.
2. **Third-party credibility:** partnering with **Conservation International** removed the "greenwashing discount" and solved the information asymmetry.
3. **Value capture: premium pricing** absorbs the cost of C.A.F.E. standards without sacrificing operating margins.

MODULE 5 · CLASS 2 SLIDES (18 OCT 2026)

Beyond the Compliance Mirage

Local fractures, shared river basins and transboundary environmental risk · Plachimada · Flint · Ostrom CPRs · Cauvery · Indus, Brahmaputra, Mekong, Nile

5.1 The provocation: water scarcity as the immediate ESG frontier

EXPLAINED SIMPLY

Carbon is a slow, global problem. Water is a fast, local one. When a local water source runs dry, a factory stops tomorrow, so water is the most immediate ESG risk.

> **Example:** Chennai's 'Day Zero' water crisis in 2019 forced offices to ask staff to work from home and hotels to ration water. That had nothing to do with carbon prices.

- **Immediacy over climate:** carbon dominates headlines, but water depletion is a **tangible, local and immediate** threat to operations.
- **Supply–demand deficit:** global freshwater demand is outpacing sustainable replenishment, driven by urbanisation, population growth and industrial and agricultural intensification.
- **The core conflict:** water is **fixed and hyper-local**, which sets up inevitable friction between industrial users and host communities.

Economic framing: human right vs. economic good

EXPLAINED SIMPLY

Water has two roles. A small amount (25–50 litres per person per day) is a basic right that should never be denied. Beyond that it is an economic good that should be priced so it isn't wasted.

> **Example:** Drinking water for a family is a right. Water for a golf course, a car wash or a bottling plant is a commercial use that should be paid for properly.

1. Basic human right	2. Economic good
25–50 litres per person per day for drinking and basic sanitation. A non-negotiable social entitlement and the **fiduciary obligation of the state**.	Industrial cooling, bottling, lawn irrigation, luxury use. **Zero pricing causes over-extraction**, so price signals are needed for **allocative efficiency**.

THE POLITICAL-ECONOMY PARADOX

When states fail to separate these two uses, firms become targets of community resistance over what looks like **privatisation of a common-pool resource**.

Supply chains: the concealed water footprint

EXPLAINED SIMPLY

Most of a company's water use is hidden in its suppliers, especially farms. Saving water inside the factory isn't enough if suppliers use it wastefully.

Example: A cotton T-shirt maker may save 10% of water in its dyeing unit, but about 2,700 litres were already used to grow the cotton on farms. The bigger fix is helping farmers switch to drip irrigation.

- **Direct vs. indirect:** factories are a small fraction of a company's water footprint. **Over 70% of global freshwater extraction** is tied to agriculture (upstream suppliers).
- **Systemic ESG vulnerability:** efficiency inside the plant wall is not enough. Inefficient upstream irrigation (flood instead of micro-irrigation) destabilises the whole watershed.
- **Policy distortions:** heavy subsidies on electricity and water hide real scarcity and discourage conservation.

Corporate strategy meets local governance

EXPLAINED SIMPLY

A company may hold a licence from the state capital, but the water is pumped in a village. If the village council and people object, they can stop the operation. A legal permit isn't the same as local acceptance.

Example: A sand-mining firm with a state licence still gets blocked when the gram sabha passes a resolution against it and villagers block the trucks.

- **Limits of central permits:** big firms get licences from national or state ministries, but the water is physically pumped at the **village or municipal aquifer**.
- **The local governance veto:** local councils, Panchayats and municipal boards have **democratic legitimacy**. When wells run dry they revoke licences and mobilise resistance.
- **Lesson:** formal state legality ≠ local societal legitimacy, also called the **Social Licence to Operate**.

5.2 The compliance mirage

EXPLAINED SIMPLY

A 'mirage' is something that looks real from far away but isn't. Firms think 'we meet all global carbon and reporting rules, so we're safe', but local problems (water, community anger) can still shut them down.

Example: A student who submits every assignment on time thinks the year is safe, then gets barred from exams for low attendance. One set of rules was met, but another was ignored.

The standard executive playbook: (1) decarbonise, (2) publish double-materiality reports (CSRD/ISSB), (3) achieve top-decile ESG ratings, and conclude: "Once we are border-tax compliant, our capital is de-

risked."

DEFINITION: THE COMPLIANCE MIRAGE

The belief that meeting global carbon and disclosure frameworks protects a firm's cash flows from **local environmental and institutional risk**.

BOARDROOM STRESS TEST: THE €600M LOW-CARBON PLANT

Day 0: everything is right	Month 6: the fracture	Financial consequence
Fully CBAM-compliant with excellent carbon intensity; long-term EU OEM supply contracts signed; equity underwriters satisfied.	Village council refuses permit renewal; farmers blockade the logistics gate; drought, and the state water authority steps in.	Utilisation falls **95% → 0%**; Debt-Service Coverage Ratio (DSCR) collapses; technical debt default.

Why does the DSCR collapse? DSCR = cash available for debt service ÷ debt payments due. Here is a simple illustration of the slide's logic:

1. Suppose that at 95% utilisation the plant earns operating cash of €95M a year and owes €60M a year in debt service. DSCR = 95 ÷ 60 ≈ **1.58**, which is healthy (lenders usually require more than about 1.2–1.3).
2. Operating cash scales with utilisation, and fixed costs keep running. At 0% utilisation, operating cash ≈ €0 or negative. DSCR = 0 ÷ 60 = **0**.
3. DSCR falls below the loan covenant, which triggers a **technical default** even though the plant is "perfectly compliant".

The paradox: a perfect CBAM certificate in Brussels, but factory gates chained shut in the sub-district.

Two ledgers, two governance scales

EXPLAINED SIMPLY

There are two account books: a global one for carbon (any ton saved anywhere counts) and a local one for water, soil and wildlife (only local savings help locally).

Example: Planting trees in Brazil can offset carbon emitted in Pune. But if your factory dries up a well in Pune, nothing done in Brazil can refill it.

Global ledger	Local ledger
CBAM, EU ETS, CSRD/ISSB. Governs **trade standards**. CO_2 is globally mixed and **fungible**: a ton abated anywhere counts.	Aquifers, soil, effluents, biodiversity. Governs the **physical tap**. **Non-fungible and place-bound.**

"You cannot offset a drained aquifer with a carbon credit."

"FUNGIBLE" IN SIMPLE WORDS

Money is fungible: one ₹100 note is as good as another. Carbon is similar: cutting a ton in Brazil helps the climate as much as cutting one in Pune. Water is **not** fungible: water saved in Brazil does nothing for a dry well in Palakkad.

Why it fails systematically: the nested systems model revisited

EXPLAINED SIMPLY

Carbon certificates and trade rules live in the 'economy' ring, the smallest inner circle. Water and soil live in the outer rings that everything depends on. Paperwork in the inner ring can't protect you from problems in the outer rings.

Example: Having a valid driving licence (paperwork) doesn't help if the road ahead is washed away by a flood (physical reality).

- CBAM, ETS and compliance "passports" are **financial and trade constructs** in the innermost ring (the economy).
- Water, soil and biodiversity live in the **outer rings** (society and biosphere).
- Natural capital is **non-fungible, physically bound and local**.
- A capital strategy that optimises only for the border tariff ignores the ring that controls survival.

COLD CALL: THE 40% PRECIPITATION DROP (MODEL ANSWER)

Q1. What sequence of regulatory actions follows? The water authority declares a drought, imposes extraction limits and rationing, and gives drinking water priority. Industrial permits are then suspended or curtailed. Local bodies may refuse renewals and communities may protest.
Q2. Does the state protect your production line or community drinking water? Community drinking water, every time. Under the allocation hierarchy, industry comes last.
Synthesis: statutory compliance gives you a **legal paper permit**. It does not guarantee your physical water supply when shortages strike.

5.3 Part 2: Local governance and the Social Licence to Operate

EXPLAINED SIMPLY

A **de jure** permit is legal permission from the government. A **de facto** social licence is the local community's acceptance. You need both, and losing the second can strand your investment.

Example: A mobile tower may have municipal approval, but if residents fear radiation and protest, it may never be switched on.

Statutory permit (de jure, "by law")		Social licence (de facto, "in practice")
State or central clearance: Factories Act, Water Board	≠	Community trust and resource security: aquifer level, livelihoods, health

When the two separate, the result is **governance decoupling and capital stranding**.

Case A: Plachimada, Kerala (Hindustan Coca-Cola Beverages, Palakkad)

EXPLAINED SIMPLY

Coca-Cola had every government clearance, but its plant pumped so much groundwater that local wells dried up and turned salty. The village Panchayat refused to renew its licence, courts backed the community, and the plant closed for good.

> **Example:** Imagine a neighbour with a legal borewell pumping day and night until every other house's tap runs dry. However legal it was, the neighbourhood will unite to stop him.

The setup	Within about 24 months, communities reported
Plant set up **1999–2000** on a **34-acre** plot. All state clearances in hand: factory licence, Pollution Control Board NOC, investment approvals. Estimated draw: about **500,000 litres/day** of groundwater from deep borewells, in Kerala's rice-growing corridor.	1. **Water-table depletion:** shallow wells and borewells dried up. 2. **Quality deterioration:** brackish, excessively hard water. 3. **Contaminated by-product:** effluent sludge handed out as "fertiliser"; independent tests found **cadmium and lead**.

Timeline: 1999–2000 plant commissioned with state clearances → years 1–2 wells dry up and quality declines → **April 2003 Perumatty Panchayat refuses licence renewal** → **2004 plant permanently shut down**.

- The threat came **not from national inspectors but from decentralised democratic institutions**.
- **73rd Constitutional Amendment:** Panchayati Raj bodies hold authority over sanitation, public health and local common-pool assets.
- **The company's argument:** land ownership plus Pollution Board clearance gave it the right to pump percolating groundwater.

LEGAL TURNING POINT: THE PUBLIC TRUST DOCTRINE (PERUMATTY GRAMA PANCHAYAT V. STATE OF KERALA)

- The State is **not the absolute owner** of natural resources; it is a **trustee for the community**.
- Groundwater is a **common-pool resource belonging to the public**.
- Excessive commercial extraction that deprives citizens of drinking water engages fundamental rights under **Article 21** (right to life).

Balance-sheet outcome: 100% asset stranding of the plant, plus lasting damage to brand equity that spread to sales nationwide.

PUTTING PLACHIMADA'S NUMBERS IN PERSPECTIVE

1. **Annual extraction:** 500,000 L/day × 365 = **182.5 million litres a year**.

2. **Equivalent basic human need** (using the 25–50 L/person/day "human right" range):

```
500,000 ÷ 50 = 10,000 people   |   500,000 ÷ 25 = 20,000 people
```

One plant's daily draw equalled the basic drinking and sanitation needs of 10,000–20,000 people, which explains why the community saw it as a threat to a human right.

Theory: Ostrom's Common-Pool Resources (CPRs)

EXPLAINED SIMPLY

A common-pool resource is shared and gets used up (rival), but it's hard to keep people out (non-excludable). Ostrom showed that local communities can manage such resources well through their own rules.

Example: A village pond: if one family takes too many fish, fewer remain for others, and it's hard to fence. Many villages have managed ponds for centuries with simple rules on fishing seasons, net sizes and who may fish.

Core idea	Executive reading
CPRs (aquifers, fisheries, forests) are **rival** but it is **hard to exclude** users. Neither pure privatisation nor pure state control is the only answer. Communities can govern CPRs sustainably through local, legitimate rules.	A firm drawing on a CPR is a **participant in a commons**, not just a permit holder. Ignoring the commons' governing institutions invites collective pushback. **Legitimacy is earned with local bodies**, not only with state regulators.

Case B: Flint, Michigan (USA)

EXPLAINED SIMPLY

To save $5 million, unelected officials switched Flint's water source to a corrosive river and skipped a cheap $100–150/day chemical treatment. Lead leached from old pipes into homes, poisoning children. The final cost exceeded $600 million.

Example: It's like saving ₹100 by skipping an oil change and ending up with a seized engine that costs ₹1 lakh. Short-term savings created a far bigger long-term disaster.

The catalyst: fiscal austerity meets centralised takeover

- A de-industrialised city with acute budget deficits. Michigan invoked emergency-management laws, displacing the elected council. Power was concentrated in an **unelected Emergency Financial Manager**.
- Flint had bought treated **Lake Huron** water via Detroit (DWSD). In **April 2014** it switched temporarily to the **Flint River**, pending a new regional pipeline (KWA). **Goal: save about $5 million over two years.**
- **Core governance failure:** essential public goods were subordinated to short-term fiscal metrics, cutting off accountability to citizens.

The biophysical failure: missing corrosion control

- Flint River water had much **higher chloride**, was more acidic and corrosive to old pipes.
- Federal rules require treating corrosive water with **orthophosphate** to protect lead pipes, at an estimated **$100–$150 per day**. The state regulator (MDEQ) misread the rules and skipped it.
- The acidic water stripped the protective mineral crust inside old lead service lines, and **lead leached into drinking water**.

Institutional denial

- Red flags in 2014–15: brown water, odours, rashes, hair loss; boil-water alerts for **E. coli**; a **Legionnaires' disease** outbreak with at least 12 deaths.
- Agencies insisted the water met standards; sampling was manipulated (e.g., **pre-flushing taps**); complaints were dismissed as "hysteria".
- **Lesson:** compliance on paper can completely decouple from real-world safety.

The aftermath

- **Irreversible health damage:** lead is a neurotoxin with **no safe threshold for children**, causing permanent cognitive and behavioural damage and chronic heart and kidney disease.
- **Infrastructure destruction:** damaged pipes kept shedding lead even after the switch back, forcing citywide pipe replacement.
- **Economic inefficiency:** an effort to save $5M led to **over $600M in legal settlements**, billions in health and remediation costs, and the total loss of municipal credibility.

NUMERICAL – THE FLINT "FALSE ECONOMY"

1. **Cost of corrosion control skipped** over two years:

```
Low:  $100/day × 730 days = $73,000
High: $150/day × 730 days = $109,500
```

2. **Planned saving** from switching = $5,000,000. Even after paying for orthophosphate, the net saving would still have been ≈ $4.89–4.93 M.
3. **Actual cost** = settlements alone over $600 M. Ratio = 600 ÷ 5 = **120×** the intended saving, before counting healthcare and pipe replacement.

Skipping a ~$100–150/day treatment to protect a $5M saving produced over $600M in liabilities, at least 120× the "saving". This is the downstream tail risk of short-term cost minimisation.

CORE LESSONS FOR FIRM-LEVEL ESG AND GOVERNANCE

1. **The flaw of short-term cost minimisation:** ignoring ecological and chemical realities creates tail risks far bigger than the initial savings.
2. **Environmental justice:** marginalised communities absorb negative externalities disproportionately when accountability is removed.
3. **The fragility of the social licence:** once transparency and trust collapse, recovery takes generations.

Comparative matrix: local governance failure

EXPLAINED SIMPLY

Both cases show 'paper compliance' separating from reality. In Plachimada a private firm ignored the community; in Flint the state ignored its own citizens. In both, local people paid the price and later forced change.

Example: In both cases the official file said 'all clear' while people's taps said otherwise.

Dimension	Plachimada, Kerala	Flint, Michigan
Institutional collision	Private extraction vs. common-pool community base	Fiscal austerity vs. public-health infrastructure
Governance tier conflict	Panchayat (73rd Amendment) vs. State Pollution Control Board	State Emergency Manager vs. elected municipal council
Regulatory blind spot	Groundwater treated as a private property entitlement	Neglect of water chemistry (corrosion inhibitors)
Balance-sheet outcome	100% asset stranding; loss of social licence	$600M+ settlements; ongoing infrastructure recapitalisation

COMMON THREAD

When institutions treat natural capital (K_n) as an **unpriced externality**, deferred legal, capital and social liabilities can paralyse operations.

DISCUSSION PROMPT ANSWERED: FLINT VS. PLACHIMADA

Both show legal or bureaucratic "compliance" separating from community reality. In Plachimada a **private firm** held valid clearances but bypassed community rights over a commons. In Flint a **public authority** bypassed elected local government to save money. In both, those closest to the resource (Panchayat, citizens) bore the harm and eventually forced the reckoning, through a licence refusal and courts in one case and lawsuits and public outcry in the other.

5.4 Part 3: Regional ecosystems and inter-state coordination (the Cauvery dispute)

EXPLAINED SIMPLY

Rivers ignore state borders. The upstream state (Karnataka) controls the water first; the downstream state (Tamil Nadu) depends on what's released. In good monsoon years there is enough; in drought years both need more than exists, and the conflict flares.

Example: Two flats share one overhead tank. The upper flat fills buckets first. In summer, when the tank is half-full, the lower flat gets almost nothing and a fight breaks out.

- **Upstream: Karnataka.** Bengaluru's urban core and Mandya (sugarcane, paddy). Faces monsoon volatility and reduced runoff.
- **Downstream: Tamil Nadu.** The Cauvery Delta (Kuruvai/Thaladi crops). Faces deficit transfer and crop-failure risk.
- **Ecosystems don't follow administrative borders:** upstream extraction changes downstream base flows, sediment balance and aquifer recharge.
- In surplus monsoons the system works. In a **30–40% Southwest Monsoon deficit it gridlocks**, because release rules are fixed monthly volumes.

Karnataka (upper riparian)	Tamil Nadu (lower riparian)
Rapid urban-industrial growth; Bengaluru's water is pumped from the basin over long distances and heights. Water-intensive command areas in Mandya. **Drought trade-off:** release water to meet downstream quotas, or protect its own municipal and farm needs?	Delta farming depends on multi-crop cycles and steady flows via **Mettur Dam**. Flows also **prevent seawater ingress** and recharge groundwater. Restricted outflows mean crop distress and falling water tables.

WHY FIXED MONTHLY VOLUMES GRIDLOCK IN A DEFICIT (ILLUSTRATION)

1. Suppose a normal month brings 100 units of inflow at the upstream dam. The award requires 40 units released downstream, leaving 60 for upstream use.
2. In a 35% monsoon deficit, inflow = 100 × (1 − 0.35) = 65 units. A fixed release of 40 leaves upstream only 25 units, a **58% cut** ((60 − 25) ÷ 60) for upstream users versus **0%** for downstream.
3. A **proportional (distress-sharing) rule** would cut both by 35%: downstream gets 40 × 0.65 = 26, upstream gets 60 × 0.65 = 39.

Fixed volumes push all the shortage onto one party, which is why the system works in surplus years but breaks down in deficit years.

National Water Policy (2012): IWRM and the allocation hierarchy

EXPLAINED SIMPLY

Plan water for the whole river basin, not state by state. In shortages, the order is drinking water first, then food and livelihoods, then nature's minimum flows, and industry last.

Example: In a summer shortage, a city cuts supply to factories and construction sites before cutting household drinking water. Industry always waits at the back of the queue.

IWRM PRINCIPLE (INTEGRATED WATER RESOURCES MANAGEMENT)

The **entire river basin or sub-basin** is the planning unit, not the district or state.

Allocation priority (first served → last served):

1. Safe drinking water and sanitation (pre-emptive need)
2. Food security and livelihoods
3. Ecological needs (minimum river flows)
4. **Commercial and industrial use**: **curtailed first** under stress

Executive implication: industry sits at the bottom. In a stressed, contested basin, commercial allocations are cut first, so assuming uninterrupted intake is an **unhedged operational risk**.

5.5 Part 4: Transboundary hydrological risk and geopolitics

EXPLAINED SIMPLY

When a river crosses *countries*, there's no common court to settle disputes; it becomes diplomacy. The upstream country has physical control and can build dams or hold back water.

> **Example:** China builds dams on the Brahmaputra upstream, and India and Bangladesh downstream worry about flow and floods but can't simply sue China.

- Across sovereign borders there is **no supreme court or central authority** to arbitrate, so governance shifts to **geopolitics and treaty negotiation**.
- The recurring structure is **upper riparian vs. lower riparian**.

Basin	Riparian states
Indus	India, Pakistan
Brahmaputra	China, India, Bangladesh
Mekong	China, Myanmar, Laos, Thailand, Cambodia, Vietnam
Nile	Ethiopia, Sudan, Egypt

THE HARMON DOCTRINE

Absolute territorial sovereignty: a state may dam, divert or consume waters that start in its territory regardless of harm downstream. **Rejected by modern international law.** Yet upstream states still have **physical control** through run-of-river cascades, inter-basin diversions, storage reservoirs, holding back lean-season flows, and sudden unannounced flood releases that cause flash floods and siltation.

International water law: three core principles

EXPLAINED SIMPLY

Three basic rules: share fairly (which doesn't mean 50–50), don't cause serious harm to others, and warn neighbours and share data.

Example: In a joint family kitchen, everyone gets a fair portion (more for those who need more), nobody spoils the food for others, and you tell everyone before using up the last of the rice.

Helsinki Rules (1966) and the **UN Watercourses Convention (1997)**:

1. **Equitable and reasonable utilisation:** each riparian has a right to an equitable share. **Equitable ≠ 50–50.** It is judged by population dependence, alternatives, economic need and historical use.
2. **Obligation not to cause significant harm:** due diligence and transboundary environmental impact assessments (EIAs).
3. **Prior notification and data sharing:** exchange of hydrological data, streamflow and flood warnings in near real time.

Operational reality: treaties depend on trust and transparency

EXPLAINED SIMPLY

Treaties work only if countries trust each other and share data. In tense times, upstream countries may hide dam levels as 'secret', leaving downstream businesses in the dark.

Example: A factory below a dam suddenly floods because water was released at night without warning. Nobody had told it the data.

- In geopolitical standoffs, upstream states may classify basic telemetry (dam levels, silt loads, discharge rates) as **national-security data**.
- **Exposure for downstream assets:** plants, logistics hubs and energy projects sit below infrastructure they have no real-time visibility into. An unannounced diversion or release can change your operating environment overnight.
- **Scaling up the pattern:** Plant → Panchayat → State → Nation → International basin. At each tier a **different institution controls the tap**.

5.6 Part 5: Executive capstone, the water-stressed greenfield

EXPLAINED SIMPLY

The question here is how to save a 'perfect on paper' plant hit by drought and community anger, and what should have been done before building it.

Example: Fixing it now = cut pumping, share water, recycle, talk to the Panchayat. Preventing it = study the aquifer, talk to villagers, and design for zero groundwater dependence *before* spending the money.

The investment	Year 2: severe monsoon drought
€400M greenfield advanced-materials plant. Fully CBAM-compliant (induction furnaces, long-term renewable PPAs). Top-quartile ESG rating; ISSB/CSRD disclosures satisfied. Semi-arid, fast-growing industrial corridor; basin shared by two states. Signed permits: **600,000 L/day** of groundwater from on-site borewells.	Farm lobbies demand a complete shut-off of non-essential industrial intake. A village council sit-in at your gates; nearby domestic wells have dried up. Local media focuses on your extraction while farmers face scarcity.

Q1. IMMEDIATE CRISIS CONTINUITY: HOW DO YOU AVOID A TOTAL SHUTDOWN WITHOUT ILLICIT WATER TRUCKING?

- **Voluntarily cut extraction** at once and publish real-time meter data on a public dashboard (transparency rebuilds trust).
- Switch to **treated municipal wastewater / recycled water**, raise in-plant recycling (closed loops, ZLD), and run reduced shifts that prioritise the most water-efficient product lines.
- **Share water with the community:** supply drinking water to affected villages from treated sources, and fund well deepening or recharge.
- Negotiate with the Panchayat and water-users' association a temporary drought-sharing agreement with agreed thresholds. Do not lean only on the state permit.

Context: 600,000 L/day ÷ 50 L ≈ the basic daily water need of 12,000 people (24,000 at 25 L/person), which is why the optics are so damaging.

Q2. PRE-INVESTMENT GOVERNANCE ARCHITECTURE: WHAT SHOULD HAVE BEEN DONE BEFORE COMMITTING CAPITAL?

- **Basin-level water-stress test:** hydrological study of the aquifer and basin, including 30–40% monsoon-deficit scenarios and inter-state allocation risk.
- **Free, prior and informed engagement** with Panchayats and farmer groups, and a formal community benefit-sharing agreement (the social licence before the statutory permit).
- **Design for resilience:** closed-loop and zero-liquid-discharge systems, rainwater harvesting, recycled-water sourcing; avoid depending on groundwater.
- Join or create a **water-users' association** with joint monitoring, following Ostrom's principles.
- Build drought curtailment into financial models and loan covenants (DSCR stress tests), and consider insurance or contingency reserves.

KEY TAKEAWAYS (CLASS 2)

1. **Compliance ≠ legitimacy.** A statutory permit is not a social licence.
2. **Carbon is fungible; water is not.** Global frameworks cannot offset local depletion.
3. **Governance is multi-tiered.** Panchayat, state, nation and international basin each control a different part of the tap.
4. **Industry ranks last** in the water allocation hierarchy under stress.
5. **Design for resilience:** closed loops, community engagement, and basin-level stress tests before CapEx.

Closing thought: meeting CBAM is the cost of market access; true resilience is built from the ground up.

MODULE 6 · SLIDES (19 OCT 2026)

Payments for Ecosystem Services (PES)

From market failure to Coasean bargaining · pricing · ecosystem services · Wunder's criteria · upstream–downstream game · irrigation tanks · Deepwater Horizon

6.1 Why prices matter: three core functions

EXPLAINED SIMPLY

Prices do three jobs: they tell us what's scarce, they encourage us to save or produce more, and they decide who gets limited goods.

Example: When onion prices jump to ₹100/kg: buyers use fewer onions (signal), farmers plant more next season (incentive), and onions go to those who value them most (rationing). No government order is needed.

1. Signal	2. Incentive	3. Rationing
Prices tell producers and consumers how **scarce** a resource is. A rising price signals scarcity.	Higher prices reward conservation, substitution and innovation. Lower prices encourage use.	Scarce goods go to those who value them most, without central planning.

KEY IDEA (HAYEK, 1945)

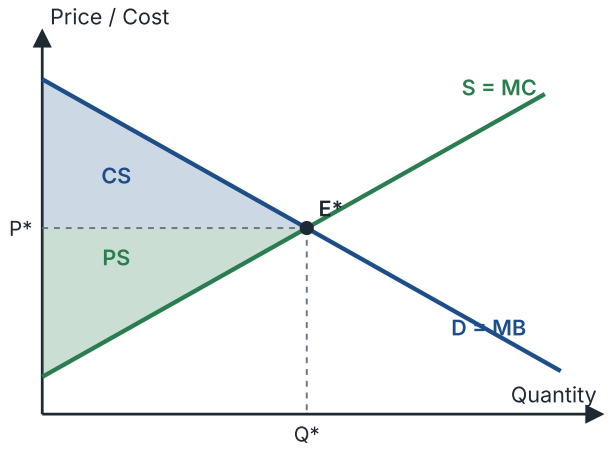
A price is a compact carrier of information: it sums up the preferences and costs of millions of people. When tomatoes get expensive, everyone knows to use fewer, without anyone giving an order.

Efficient allocation: equilibrium and surplus

EXPLAINED SIMPLY

Where supply meets demand, everyone who values a good more than it costs gets it. That's the point of maximum total benefit for buyers and sellers together.

Example: At a weekly market, mangoes settle at ₹80/kg. Anyone willing to pay ₹80 or more buys, and any grower who can sell profitably at ₹80 sells. Nothing is wasted.



Competitive equilibrium: total surplus is maximised

- D reflects **marginal benefit** (willingness to pay); S reflects **marginal cost**.
- The market clears at P*, Q* where **MB = MC**.
- **Total surplus (CS + PS)** is maximised there. Any other quantity lowers total welfare.

When does pricing deliver efficiency?

EXPLAINED SIMPLY

Markets work perfectly only in ideal conditions: many sellers, clear ownership, everyone well informed, no side-effects on others, private goods only, and every useful thing has a price. Real life rarely meets all six.

Example: A vegetable market comes close. The 'market' for clean air fails almost every condition: no owner, no price, and everyone breathes it.

First Fundamental Theorem of Welfare Economics: a competitive equilibrium is **Pareto efficient** (no one can be made better off without making someone worse off), provided:

1. Many buyers and sellers (perfect competition, no market power)
2. Well-defined, enforceable property rights
3. Full and symmetric information
4. No externalities: private cost = social cost
5. All goods are private goods (rival and excludable)
6. Complete markets: every valuable good has a price

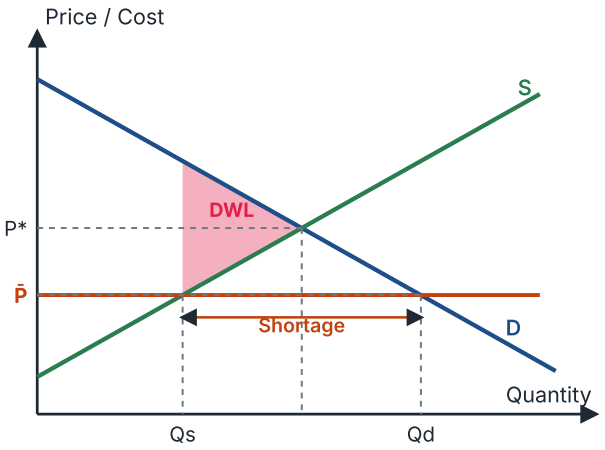
If **any** condition fails, the price no longer reflects true social value.

Government controls (1): price ceiling

EXPLAINED SIMPLY

A price ceiling is a legal maximum price. If it's set below the market price, people want more than sellers will supply, which causes shortages, queues and black markets.

Example: Very cheap or free farm electricity means farmers pump groundwater non-stop, because the cost of using more is almost zero. That is a big reason water tables are falling in Punjab.



Binding price ceiling: shortage + deadweight loss

- A binding ceiling $\bar{P} < P^*$ causes $Q_d > Q_s$, a **shortage**.
- Trade falls to Q_s, and a **deadweight loss** appears.
- This leads to queues, black markets and declining quality.
- **Environmental angle:** artificially cheap water or power encourages over-use.

Government controls (2): floors, taxes and subsidies

EXPLAINED SIMPLY

Support prices, subsidies and taxes push prices away from real costs. When the price is wrong, resources are misused, often harming nature.

Example: A high MSP for paddy encourages farmers in dry regions to grow water-hungry rice instead of millets, using up scarce groundwater.

Intervention	Market effect	Typical example
Price floor (support price)	Surplus; resources locked into the supported activity	Minimum support price encouraging water-intensive crops
Input subsidy	Price below marginal cost, so over-use	Free or cheap electricity for pumping; fertiliser subsidy
Output tax / quota	Wedge between buyer and seller price; DWL if set wrongly	Excise duties, export bans

TAKEAWAY

Controls drive a **wedge between price and marginal cost**. Allocation moves away from the efficient point, and the resource is often degraded through over-use. *Note:* intervention is not always bad. A corrective tax can fix an externality. The problem is intervention that distorts prices **without a welfare rationale**.

6.2 Why pricing fails

EXPLAINED SIMPLY

Prices fail to protect nature because many natural benefits have no owner, no market and no price, so in business decisions they count as zero.

Example: A forest's timber has a price, but the rain it attracts and the floods it prevents don't. So clearing the forest looks profitable.

(1) The nature of the good

EXPLAINED SIMPLY

Two questions decide what kind of good something is: does my use reduce yours (rival)? And can people be kept out (excludable)? Public goods and common-pool resources are hard to exclude, so they get overused or under-supplied.

Example: A street light (public good) helps everyone, so nobody wants to pay for it. A village well (common pool) is open to all, so everyone draws too much.

	Excludable	Non-excludable
Rival	**Private good** timber, bottled water	**Common-pool resource** fisheries, groundwater
Non-rival	**Club good** national park with entry fee	**Public good** climate stability, clean air

- **Public goods:** non-excludability leads to **free-riding**, so no one pays and the market under-supplies them.
- **Common-pool resources:** each user ignores the cost imposed on others, which produces the **tragedy of the commons** (Hardin, 1968).

(2) Missing markets and property rights

EXPLAINED SIMPLY

If no one owns a resource and there's no market for its services, nobody is paid to protect it. High bargaining costs and a focus on the short term make things worse.

Example: Bees pollinate a farmer's crops for free. No bee 'market' exists, so when pesticides kill the bees, no price signal warns anyone.

- **Missing markets:** many ecosystem services (carbon storage, water purification, pollination, flood control) have **no price**, so decisions value them at **zero**.
- **Weak property rights:** when no one owns or can enforce rights over a resource, no one has an incentive to conserve it. **Coase (1960):** clear rights plus low transaction costs could resolve conflicts

through **bargaining**.

- **High transaction costs:** many beneficiaries, many providers and difficult monitoring block private bargaining.
- **Time and uncertainty:** future and irreversible losses are heavily discounted, so long-run ecological value is under-priced.

RESULT

Market price < true social value, which leads to **over-exploitation and ecological degradation**.

BRIDGE TO PES: THE CENTRAL QUESTION

Can PES **create the missing price signal** so that those who conserve nature are rewarded and degradation falls? Sub-questions: *Who pays and who provides? How is the payment set? Does it work without clear rights and monitoring?*

6.3 Ecosystem services: the economic dimension

EXPLAINED SIMPLY

Ecosystem services are the free benefits nature gives us: water, food, clean air, flood control, pollination and recreation. Because they're free, we overuse them, and more than 60% are being worn out faster than they recover.

Example: Mangroves in the Sundarbans protect villages from cyclones for free. Building a concrete sea wall to do the same job would cost crores.

- **Definition:** the flow of benefits and value that societies and economic agents get, directly or indirectly, from healthy natural capital.
- Most ecosystem services are **public goods** or **common-pool resources**. With zero explicit prices they suffer continuous over-extraction, under-provision and negative externalities.
- **Global status:** over **60%** of global ecosystem services are being degraded faster than they can regenerate.

Typology of ecosystem services (CICES)

EXPLAINED SIMPLY

Three types: **provisioning** (things we take: food, timber, water), **regulating** (services that keep nature balanced: water filtering, flood control, carbon capture) and **cultural** (non-material: beauty, tourism, sacred sites).

Example: A forest gives timber and honey (provisioning), stops landslides and cleans the stream (regulating), and hosts a temple and trekking trails (cultural).

1. Provisioning	2. Regulating	3. Cultural
Direct material and energy inputs: timber, fibre, food, freshwater, biomass fuel. *Usually traditional private goods with explicit market prices.*	Keeping the biophysical balance: filtration, waste breakdown, flood defence, carbon capture. *Positive externalities with no price discovery.*	Non-material direct benefits: aesthetic value, ecotourism, hunting, sacred heritage. *Mixed club/public goods.*

THE CORE POLICY PARADOX

Markets reward **provisioning** conversions (clear-cutting forests for timber) while ignoring the collapse of **regulating** services, because only the timber has a price.

6.4 PES theory: Sven Wunder's five criteria

EXPLAINED SIMPLY

PES means 'pay the people who protect nature for the service they provide'. It must be voluntary, the service must be clearly defined, there must be a real buyer and a real seller, and payment happens only if the service is delivered.

Example: A city water board pays upstream farmers ₹X per acre per year to keep trees along the river, and stops paying if the trees are cut. That is all five criteria in one deal.

FORMAL DEFINITION (WUNDER, 2005)

A **voluntary transaction** where a **well-defined ecosystem service** (or a land use likely to secure it) is **bought by at least one buyer** from **at least one provider**, **conditional** on continued provision.

1. **Voluntary participation:** neither party is forced by law.
2. **Well-defined ecosystem service:** clear metrics for the service or a proxy land use.
3. **Direct beneficiary payment:** at least one real buyer funds it.
4. **Ecosystem stewardship:** at least one land or resource manager supplies it.
5. **Conditionality:** payment happens **if and only if** the service is sustained.

Memory aid: **V-W-B-S-C**, "Voluntary, Well-defined, Buyer, Seller, Conditional".

Economic pricing: the opportunity-cost principle

EXPLAINED SIMPLY

A landowner will protect nature only if the payment is at least what they would earn by using the land another way (WTA). The buyer will pay at most what the service is worth to them (WTP). A deal is possible when WTP ≥ payment ≥ WTA.

Example: A farmer would earn ₹50,000 a year by clearing land, so he needs at least ₹50,000 to conserve it. A bottling company would spend ₹2 lakh on filters otherwise, so it can pay up to ₹2 lakh. Any payment between ₹50,000 and ₹2 lakh works for both.

- **Land manager's trade-off:** choosing conservation means giving up profits from intensive farming, forestry or development.
- **WTA (willingness to accept):** the landowner's **minimum** payment = **net forgone profit** from the alternative land use (the opportunity cost).
- **WTP (willingness to pay):** the beneficiary's **maximum** payment = the value of the service to them (e.g., the cost they avoid).

CONDITION FOR A FEASIBLE CONTRACT

$$WTP_{beneficiary} \geq Payment \geq WTA_{provider}$$

If payment < opportunity cost, the landowner defects. If payment > marginal benefit, the buyer exits.

6.5 Types of PES with examples

EXPLAINED SIMPLY

PES can be paid for by government (taxpayer money), by private companies that benefit directly, or by interest groups such as hunters and tour operators.

Example: Government: EU pays farmers for buffer strips. **Company:** Nestlé pays Vittel farmers to keep water clean. **Interest group:** hunters pay farmers to create winter wetlands for birds.

PUBLICLY FUNDED PES: EU COMMON AGRICULTURAL POLICY (CAP)

- Mechanism: **agri-environment-climate measures (AECM)** under CAP Pillar II.
- Cost share: the **European Commission pays up to 75%** of marginal costs and **member states/regions pay the remaining 25%**.
- Schemes: **Sachsen-Anhalt (Germany)** compensates integrated crop management, cover crops and wild-meadow strips. **Denmark and France** compensate buffer strips along waterways to cut nitrogen run-off and restore ecological connectivity.
- Nature: **government-mediated** PES (taxpayer-funded, broad targeting).

Illustration: if a buffer strip costs a farmer €400/ha/year in lost output, the EU pays 0.75 × 400 = **€300** and the national/regional government pays 0.25 × 400 = **€100**.

PRIVATE USER-FINANCED PES: NESTLÉ WATERS (VITTEL, FRANCE)

- **Crisis:** intensive dairy farming in the Vosges threatened aquifer purity with pesticides and nitrates.
- **Contracts (1989–2004): 30-year** binding contracts with local farmers; **€200/ha/year for up to 7 years** to cover transition costs; Nestlé bought land and **leased it back free** to compliant farmers; financed composting facilities and equipment up to **€150,000 per farm**.
- **Economic result:** it was **cheaper** for Nestlé to pay for regenerative farming than to build an industrial water-filtration plant.

Quick maths: maximum transition payment per hectare = €200 × 7 years = **€1,400/ha**.

CO-MANAGEMENT AND CULTURAL SERVICES: HUNTING ASSOCIATIONS (EMILIA-ROMAGNA, ITALY)

Hunting groups pay farmers to **flood low-lying plots after harvest**, creating seasonal wetlands for migratory waterfowl and native birds. Farmers earn from idle winter land; hunters get habitat quality and access. **Lesson:** PES can use private interest groups to deliver wider biodiversity gains **without public spending**.

UNEP stepwise framework for operational PES

EXPLAINED SIMPLY

To set up PES properly: (1) measure the service and the farmers' costs, (2) check land rights and that contracts can be enforced, (3) design the payment plan, (4) monitor independently and withhold payment if rules are broken.

Example: Like hiring a contractor: assess the job, check the contractor is legitimate, sign a clear contract, inspect the work before paying.

Phase 1: Scoping and valuation	Phase 2: Capacity and institutions	Phase 3: Contract structuring	Phase 4: Monitoring and baseline
Define the spatial and hydrological boundaries of the service. Quantify the baseline ecosystem status. Calculate providers' opportunity costs.	Validate tenure and property rights. Verify legal enforceability of contracts.	Transparent payment schedules (flat vs. tiered by performance). Allocate monitoring and transaction costs.	Independent third-party audit. Enforce conditionality: **withhold payments for non-compliance**.

Critical policy caveats

EXPLAINED SIMPLY

Three traps: paying for something that would have happened anyway (additionality), damage simply moving elsewhere (leakage), and protection ending when the money stops (permanence).

Example: Paying a farmer not to farm a rocky slope he'd never farm anyway is wasted money. Saving one forest while loggers move to the next valley achieves nothing overall.

- **Additionality:** are we paying for conservation that would have happened anyway?
- **Leakage:** does conservation here shift degradation to nearby land?
- **Permanence:** what happens when the funding stops?

6.6 Can PES solve environmental externalities? The core model

EXPLAINED SIMPLY

The farmer upstream bears the cost of keeping the forest, but the city downstream gets all the benefit (clean water). Without payment, the farmer clears the forest. If the city pays enough, both are better off.

Example: Bengaluru's water board paying farmers in the Cauvery catchment to keep forest cover, instead of spending far more on water treatment, is the same logic as the class numerical.

The fundamental market failure: ecosystem services (watershed filtration, carbon sinks, flood control) give vital social value but are treated as unpriced public goods. Landowners get **zero price signals** for the positive externalities they create. Command-and-control faces political resistance, and pure altruism fails against private profit. **The Coasean hypothesis:** can voluntary, conditional payments create a market and **internalise the externality**?

NUMERICAL 5 – UPSTREAM FARMER VS. DOWNSTREAM WATER UTILITY (THE CLASS PES GAME, FULLY SOLVED)

The setup

Upstream landowner (500 ha)	Downstream water utility (city)
Action 1: Conserve forest (sustainable extraction/agroforestry). Modest private return = **30**. **Action 2: Clear and intensify** (chemical monoculture/livestock). High private return = **100**. **Private opportunity cost of conserving** = 100 − 30 = **70**.	Supplies clean drinking water downstream. **If upstream conserves:** natural gravity filtration works, so net return = **80**. **If upstream clears:** natural filtration destroyed = **−120**; with an advanced treatment plant = **−60**.

The structural wedge: upstream bears 100% of the conservation cost, while downstream captures 100% of the filtration benefit. That is a **positive externality** of conservation.

Part A: Baseline game with no transfers (market failure)

Upstream ↓ / Downstream →	Natural filtration	Advanced plant
Conserve	(30, 80) ← social optimum	(30, 40)
Clear & intensify	(100, −120)	(100, −60) ← Nash equilibrium

Payoffs are written as (Upstream, Downstream).

1. **Find upstream's best strategy.** Compare its payoffs row by row in each column:

```
If downstream plays Natural: Clear 100 > Conserve 30
If downstream plays Plant:   Clear 100 > Conserve 30
```

Clear is better whatever downstream does, so **Clear is upstream's dominant strategy**.

2. **Downstream's best response** to Clear: Natural gives −120, Plant gives −60. Since −60 > −120, it **builds the plant** (minimises the loss).

3. **Nash equilibrium** = (Clear, Advanced plant) = (100, −60).

```
Welfare at Nash:     W(Nash) = 100 + (−60) = 40
```

4. **Social optimum**: the cell with the highest total payoff:

```
(30,80) → 110   (30,40) → 70   (100,−120) → −20   (100,−60) → 40
W(Optimum) = 30 + 80 = 110
```

5. **Deadweight loss:**

```
DWL = W(Optimum) − W(Nash) = 110 − 40 = 70 units of social welfare
```

Without payments, the farmer clears the land and the city builds a treatment plant: welfare = 40 vs a possible 110, so DWL = 70.

Part B: Determining the price, the Coasean bargaining range

What transfer **T** must downstream offer to make upstream conserve?

1. **Upstream's minimum price (WTA):** conserving plus payment must beat clearing:

```
Π(Conserve) + T > Π(Clear)
30 + T > 100   →   T > 70
```

Upstream rejects any offer ≤ 70 (its opportunity cost).

2. **Downstream's maximum price (WTP):** paying for clean water must beat building the plant:

```
Π(Clean) − T > Π(Advanced plant)
80 − T > −60   →   T < 140
```

Downstream refuses any offer ≥ 140.

3. **Bargaining zone: 70 < T < 140**. Any price in this range creates a mutual surplus. The total surplus available = 140 − 70 = **70** (exactly the DWL). The class contract sets **T = 80**.

Part C: Post-PES game with T = 80

Upstream ↓ / Downstream →	Offer PES (T = 80)	No offer (status quo)
Conserve	(110, 0) ← new Nash equilibrium	(30, 40)
Clear & intensify	(100, −60)	(100, −60)

(Clear, Offer) = (100, −60) because payment is **conditional**: if upstream clears, no payment is made, and the city falls back on its plant. (Conserve, No offer) = (30, 40) as in the baseline, because without a contract the city can't rely on upstream and keeps its plant.

1. **Payoffs at (Conserve, Offer):**

```
Upstream:   30 + 80 = 110   (gain of +10 over clearing at 100)
Downstream: 80 − 80 = 0     (avoids −60 loss → gain of +60)
```

2. **Upstream's response:** if offered PES, Conserve (110) > Clear (100), so it conserves. Without an offer, Clear (100) > Conserve (30).

3. **Downstream's choice (anticipating upstream):** Offer, then upstream conserves, giving 0. No offer, then upstream clears, giving −60. Since 0 > −60, it **offers the PES contract**.

4. **Efficiency check:**

```
W = 110 + 0 = 110 = social optimum   →   DWL = 0
```

The 70 surplus is split: upstream +10 (= T − 70) and downstream +60 (= 140 − T).

With T = 80 the outcome moves to (Conserve, Offer) = (110, 0). Welfare reaches the social frontier of 110, and the deadweight loss of 70 is fully eliminated. The payment is only a transfer; the efficiency gain comes from changing upstream's land use.

Extension: what transaction costs do to the deal (links to "4 structural problems")

1. Suppose monitoring, satellite verification and contract enforcement cost the utility **TC = 50**. Its WTP condition becomes 80 − T − 50 > −60, so **T < 90**.

2. New zone: **70 < T < 90**, and the surplus shrinks from 70 to 20. The class price T = 80 still works.

3. If TC ≥ 70, then T < 140 − TC ≤ 70, so there is **no feasible T** and the deal collapses even though conservation is socially efficient.

Transaction costs can "easily swallow the theoretical bargaining surplus (140 − 70 = 70)", exactly as the slide warns.

Why PES isn't a silver bullet: four structural problems

EXPLAINED SIMPLY

PES has four weak spots: people get paid for land they'd never clear anyway, damage shifts to other places, monitoring costs can eat up the gains, and paying would-be destroyers can feel unfair to people who protected nature for free.

Example: A tribal community that has protected its sacred grove for generations gets nothing, while a newcomer who threatens to cut trees gets paid to stop. That damages trust and social norms.

1. Additionality and adverse selection	2. Spatial leakage
Landowners know their opportunity costs better than buyers. Land on steep, unviable slopes (which would never be farmed anyway) enrols first, earning rent with **zero new conservation**.	If regional demand for timber or crops doesn't change, protecting this watershed just **moves logging** to the neighbouring basin.
3. High transaction costs	**4. Moral hazard and equity**
Boundary checks, satellite monitoring and contract enforcement can swallow the bargaining surplus (140 − 70 = 70).	Paying would-be clearers **rewards bad actors** while ignoring indigenous and traditional communities who conserved land for free, which erodes social norms.

6.7 Application: water transfer from irrigation tanks for urban use (India)

EXPLAINED SIMPLY

Cities are taking water from village irrigation tanks through pipelines without paying the farmers who depended on it. City dwellers gain and villagers lose.

Example: The Veeranam tank sends 180 million litres a day to Chennai. Farmers who irrigated from that tank lose water for their crops and get nothing in return.

- Rapid urban growth has worsened domestic water deficits. Instead of local water harvesting, state authorities **take rural water** for big cities.
- **Tamil Nadu example:** the **Veeranam Tank** project pipes over **180 million litres/day over 145 km** to Chennai. The **Palar** basin faces similar continuous rural-to-urban transfers.
- **Core issue:** water is taken **without compensating traditional farm users**, causing sharp rural welfare losses.

Critique of the command-and-control approach

EXPLAINED SIMPLY

When the government simply orders water to be moved, without discussion or compensation, rural people lose income, the tanks are neglected, and the city's gain becomes the village's loss.

Example: If the tank no longer serves the village, villagers stop desilting and maintaining it. Over time it fills with silt and holds less water, hurting the city too.

- **Top-down allocation:** government fiat prioritises urban drinking water without stakeholder dialogue.
- **Negative rural externalities:** lower crop yields and lost livelihoods; landless labourers, graziers and duck rearers lose post-harvest support.
- **Ecosystem degradation:** communal water bodies fall into disrepair when rural stakeholders lose the incentive to protect them.
- **Zero-sum outcome:** urban gains directly cause rural destitution.

Irrigation tanks as multi-functional ecosystem assets

EXPLAINED SIMPLY

A village tank isn't just stored water. It irrigates crops, waters cattle, raises fish and ducks, recharges wells, controls floods, and has religious value. Counting only the litres misses most of its worth.

Example: Taking the tank's water may look like 'just 1 million litres', but it also means the village wells dry up, duck farmers lose income, and the temple tank is empty during festivals.

Provisioning	Regulating	Cultural and supporting
Irrigation; livestock watering and domestic use; fish and duck rearing	Groundwater recharge; flood moderation and silt control	Sacred temple tanks and ritual use; nutrient recycling and microclimate moderation

THE PROBLEM

Conventional water diversion counts only **physical water volume** and ignores these multi-dimensional, non-market ecosystem values.

The alternative: PES for tank water

EXPLAINED SIMPLY

Instead of grabbing water, the city pays the farmers or their association for it. Farmers then have a reason to save water (drip irrigation, less thirsty crops) and keep the tank healthy, so both sides win.

Example: The city water board pays the village Water Users' Association every year. The association uses the money for desilting and drip kits and releases the saved water to the city.

A voluntary, market-based deal where the **beneficiaries** (urban water utilities) pay the **providers/stewards** (rural tank managers and farmers). It:

- Shifts policy from **coercive reallocation to contractual compensation**.
- Gives farmers an incentive to save water (drip irrigation, less water-intensive crops).
- Produces a **non-zero-sum (win-win)** outcome for farmers, city dwellers and utilities.

Global and domestic precedents of water PES

EXPLAINED SIMPLY

PES for water already works in many places: in France, Costa Rica, Mexico and South Africa, and in India at Sukhomajri (Haryana) and Rettaikulam (Tamil Nadu).

Example: At Rettaikulam, farmers charge themselves a small land-based fee (ayacut vari) to maintain their tank. Users pay for the service they get.

- **Vittel (France):** Nestlé pays dairy farmers to farm pesticide-free.

- **Costa Rica and Mexico:** hydrological payments reward forest watershed managers for downstream recharge.
- **South Africa (Working for Water):** clearing invasive catchment plants, funded by downstream consumer tariffs.
- **Sukhomajri Project (Haryana):** water and catchment sharing.
- **Rettaikulam Tank (Tamil Nadu):** farmers levy an internal **ayacut vari** (land-based water fee) to fund maintenance collectively.

PES prerequisite 1: rigorous water accounting

EXPLAINED SIMPLY

Before trading water you must count it. **Physical accounting:** how much water comes in and goes out. **Economic accounting:** what that water is worth in different uses.

Example: Like a household budget: first list income and expenses (physical), then decide which expenses matter most (economic) before cutting anything.

1. Physical accounting (necessary condition)	2. Economic accounting (sufficient condition)
Stock: beginning vs. ending basin water volume. **Flow:** inflows (rainfall, runoff) minus outflows (evapotranspiration, deep percolation, extraction). Needed to determine the **true surplus water**.	Puts monetary values on marginal gains vs. losses across sectors, using **non-market valuation** (contingent valuation, benefit transfer). Needed to justify **equitable compensation**.

WATER-BALANCE ILLUSTRATION (PHYSICAL ACCOUNTING METHOD)

A tank basin starts the season with 50 MCM (million cubic metres). Inflows: rainfall 120, runoff 30. Outflows: evapotranspiration 70, deep percolation 20, existing extraction 60.

1. Net flow = inflows − outflows = (120 + 30) − (70 + 20 + 60) = 150 − 150 = **0**.
2. Ending stock = beginning stock + net flow = 50 + 0 = **50 MCM**. The stock is stable, so there is **no true surplus**. Any urban transfer must come from savings (e.g., drip irrigation cutting farm extraction) and must be compensated.

PES prerequisite 2: property rights and local institutions

EXPLAINED SIMPLY

Farmers can only sell water they have clear rights to. Working through a Water Users' Association lets them negotiate as one group instead of thousands of individuals.

Example: A housing society negotiates one parking contract on behalf of 200 residents, which is far easier than 200 separate negotiations.

- Farmers and communities **can't bargain without formal rights** over tank water. Secure collective rights must be recognised for **Panchayats or Water Users' Associations (WUAs)**.

- **WUAs act as the collective "seller"**, negotiating directly with municipal utilities (the "buyer"). This removes heavy individual transaction costs and prevents free-riding. Revenue is distributed to members or spent on **tank desilting**.

Economic feasibility: WTP vs. WTA

EXPLAINED SIMPLY

City people already pay a lot for tanker water, so they would pay well for a reliable supply. Farmers need only to be compensated for their lost crops. The gap between the two is a 'surplus' that both sides can share.

> **Example:** If a city family pays ₹1,000 for a tanker while the farmer earns only ₹200 from that much water on his crop, paying the farmer ₹400 still saves the family ₹600, and the farmer earns double.

THE FUNDAMENTAL VALUATION CONDITION

$$WTP_{urban} \geq \text{Opportunity cost / Compensation} \geq WTA_{rural}$$

- **Urban WTP:** urban consumers and industries already pay steep private-tanker prices, so their WTP for a dependable supply is substantial.
- **Rural WTA:** compensation must fully offset forgone yields and lost local ecosystem services.
- **Efficiency gain:** water's marginal value in cities far exceeds its value in traditional farming, so a **wide trading surplus** exists.

FEASIBILITY CHECK (ILLUSTRATIVE NUMBERS, CLASS METHOD)

Suppose city households pay tankers ₹100 per kilolitre (kL), so the utility's WTP ≈ ₹100/kL. Farmers lose crop income worth ₹15/kL plus local ecosystem services (fishing, recharge) worth ₹10/kL, so WTA = ₹25/kL.

1. Check the condition: 100 ≥ Payment ≥ 25, so a deal is **feasible**.
2. Trading surplus per kL = WTP − WTA = 100 − 25 = **₹75/kL**.
3. Scale it to Veeranam's 180 million L/day = 180,000 kL/day. Daily surplus = 180,000 × 75 = **₹1.35 crore per day**, all of which command-and-control transfer leaves uncompensated to rural users.

TAKEAWAYS

Reallocating tank water through PES links **rural equity** with **urban water security**. Success needs clear property rights, empowered WUAs, and rigorous natural-capital accounting.

6.8 Case study: Deepwater Horizon through a PES lens

EXPLAINED SIMPLY

Usually PES pays people to *prevent* damage. After the BP oil spill, the same valuation methods were used in reverse: to calculate how much the polluter must pay to *repair* damage. That was over $20 billion.

Example: PES is like paying a guard to protect your house. Deepwater is like making the thief pay for everything he broke after the burglary.

Standard proactive PES	Reactive legal "PES" (Deepwater)
Voluntary transactions	Mandatory legal settlement
Beneficiaries pay resource owners	**Polluter pays** for restoration
Goal: prevent degradation	Goal: remediate catastrophic loss
Focus: maintenance incentives	Focus: compensatory valuation

Quantifying destruction: lost ecosystem services (2010 Gulf of Mexico spill)

EXPLAINED SIMPLY

The oil spill damaged all four kinds of ecosystem services at once: fishing (provisioning), coastal marshes (regulating), tourism (cultural) and marine habitats (supporting). Money can repair some of this, but not the human trauma.

Example: Fishermen lost catches, beaches closed to tourists, marshes eroded, and rig survivors suffered PTSD. The last of these can't be measured in dollars.

Category	Ecosystem service	Impact of the 2010 spill
Provisioning	Commercial fisheries	Fisheries closures, contaminated stocks
Regulating	Coastal marsh buffers	Vegetation die-off and faster erosion
Cultural	Recreation and tourism	Beach closures, lost local livelihoods
Supporting	Benthic and pelagic habitat	Widespread toxicity across marine food webs

The valuation challenge: BP's **>$20 billion** settlement was calculated with **Natural Resource Damage Assessment (NRDA)**, which relies directly on ecosystem-service valuation frameworks.

THE HUMAN TOLL: BEYOND ECONOMIC METRICS

The Guardian article "Life after Deepwater Horizon: the hidden toll of surviving disaster on an oil rig" shows that human trauma is **not captured** by market valuation:

- **PTSD and chronic stress** among rig survivors and coastal workers.
- **"Moral injury"**: a deep sense of betrayal from corporate negligence that put profit over safety.
- **Limits of PES and valuation:** money can fund wetland restoration but cannot restore psychological wellbeing. Treating nature purely as "natural capital" risks overlooking the **social, moral and human** dimensions of environmental damage.

MODULE 7 · CLASS 3 SLIDES

The Tragedy of the Commons

From Hardin's pessimism to Ostrom's design principles

7.1 Hardin's essay and the "no technical solution" claim

EXPLAINED SIMPLY

Hardin argued that some problems can't be fixed by better machines. They need people to change their behaviour and values.

Example: A filter plant can clean one polluted river (technical solution). But no machine can stop millions of people from each burning a little more fuel; that needs rules and a change in values.

In **December 1968**, ecologist **Garrett Hardin** published "The Tragedy of the Commons" in *Science*, shaping environmental economics and policy for 50 years. Most people remember the three words of the title, but almost no one remembers his **central methodological claim**:

HARDIN'S CLAIM

"The population problem has no technical solution; it requires a fundamental extension of morality."

Technical problem	"No-technical-solution" problem
A river polluted with heavy metals → an engineer builds a filtration plant → **problem solved with a machine**.	7 billion people treat the atmosphere as an open sewer → no machine can out-filter aggregate self-interest → **requires a change in values**.

"What shall we maximise?"

EXPLAINED SIMPLY

You can't maximise two competing things at once. On a finite planet, more people means fewer resources per person.

Example: A fixed-size pizza: you can make bigger slices or feed more people, not both.

- Do we maximise **population** or **resources per person**? The world is finite and can support only a finite population, so "population growth should eventually be zero."
- Geometric (compounding) population growth on a finite planet cannot be solved by engineering higher-yield wheat.
- **Bentham's maxim**, "the greatest good for the greatest number", is a **mathematical impossibility**: you cannot maximise two competing variables, **N** (number of people) and **Q** (goods per person), in one equation at the same time.

- **Conclusion:** solving the ecological crisis needs not a new machine but an **extension of morality**.

IN SIMPLE WORDS

Think of a pizza of fixed size. You can make the slices as big as possible (maximise Q) *or* feed as many people as possible (maximise N), but not both. More people means thinner slices.

7.2 The pasture model

EXPLAINED SIMPLY

When something is shared and free, each user gets the *full* gain from taking more but shares only a *small part* of the damage. So everyone keeps taking more until the resource is ruined.

> **Example:** In a shared hostel fridge, if you eat an extra piece of the common sweets you enjoy all of it, while the loss is shared among 10 roommates. Everyone thinks the same way, and the box is empty by evening.

A pasture is open to all, with a **carrying capacity of 100 cows**. There are **10 herdsmen × 10 cows each = 100 cows**, which is equilibrium. Every animal is well fed and the system is stable, until one herdsman asks: *"What happens if I add one more cow?"*

NUMERICAL 6 – THE HERDSMAN'S CALCULATION (HARDIN'S UTILITY MODEL)

1. **Positive utility:** he gets 100% of the sale value of cow #11, so private benefit ≈ **+1**.
2. **Negative utility:** the overgrazing cost of the extra cow (≈ −1) is **shared across all n herdsmen**, so his personal share is only **−1/n**.
3. **Net utility** with n = 10:

```
Net Utility = (+1) + (−1/n) = 1 − 1/10 = +0.9  > 0
```

4. **Rational conclusion:** add the cow. Note that 1 − 1/n > 0 for **any** n > 1, and the larger the group, the closer the net gain is to +1 (n = 100 gives +0.99).
5. **The tragedy:** every herdsman makes the same calculation. If all 10 add a cow, there are 110 cows on a pasture built for 100, **10% over carrying capacity**. Each cow's overgrazing cost now lands on everyone:

```
Each herdsman: gains +1 from own cow, but bears 10 × (−1/10) = −1 from all 10
extra cows
Net per herdsman = +1 − 1 = 0 this round… and the pasture is now degraded
```

Each keeps adding cows because each individual step still looks like +0.9, until the pasture collapses.

Individually rational (+0.9 each time) ≠ collectively rational (pasture ruined). "Ruin is the destination toward which all men rush, each pursuing his own best interest in a society that believes in the freedom of the commons." (Hardin, 1968)

7.3 Why not just appeal to conscience? The evolutionary double bind

EXPLAINED SIMPLY

Asking people to be good doesn't work in a free-for-all. The honest person loses out and the selfish one wins, so over time selfishness wins.

> **Example:** In a traffic jam the driver who patiently waits in line falls behind, while the one who cuts through the wrong side gets ahead. Polite requests alone don't fix the jam; traffic police and rules do.

1. You tell Herdsman 1: "Don't add another cow; think of future generations."
2. H1 listens and restrains himself.
3. H2 doesn't care and adds her 11th cow anyway.
4. H1 goes bankrupt; H2 thrives.

THE DEVASTATING IMPLICATION

An appeal to conscience **taxes the virtuous while rewarding the ruthless**. It systematically selects *against* moral restraint, which is a warning to naive corporate ESG campaigns that rely on voluntary goodwill alone.

7.4 Hardin's only two remedies

EXPLAINED SIMPLY

Hardin's two fixes were: let the government control it, or divide it up and give private ownership. Both rely on 'mutual coercion', meaning rules everyone agrees to be forced to follow.

> **Example: State:** the forest department issues permits and fines poachers. **Private:** divide a common grazing ground into fenced plots so each farmer looks after his own.

Because voluntary restraint is (in his view) an illusion, Hardin's fix is **"mutual coercion, mutually agreed upon"**:

Option A: State control	Option B: The free market (privatisation)
Central government dictates who grazes, when and how much, enforced by inspectors, fines and prison.	Fence the pasture. If H1 owns his parcel, his own cows starve if he overgrazes, so the **externality is internalised**.

DISCUSSION: TEST THE BINARY

For the atmosphere or deep-sea fisheries, would you choose a global climate dictator (state) or auction property rights to the troposphere (market)? Neither is realistic. A world government doesn't exist, and you can't fence the sky. This is the opening for Ostrom: Hardin was missing a third option, **community self-governance**.

7.5 Where Hardin went wrong

EXPLAINED SIMPLY

Hardin assumed nobody talks to anybody and there are no rules. That isn't a 'commons'; it's open access. Real communities make rules for shared resources all the time.

Example: Hardin pictured a free-for-all. A real village grazing land often has a panchayat deciding how many cattle each family may graze and when.

Hardin did not describe a "commons". He described an **unregulated, open-access free-for-all**. Goods are defined not by ideology but by two measurable properties:

- **Subtractability (rivalry):** does my use reduce what's left for you?
- **Excludability:** can I stop you from using it at all?

The 2×2 goods classification matrix

EXPLAINED SIMPLY

Goods are sorted by two questions: does using it reduce what's left (subtractability)? Can people be kept out (excludability)? That gives four types: private, club, common-pool and public.

Example: Private: a samosa. **Club:** a gym membership. **Common-pool:** groundwater. **Public:** a lighthouse, or clean air.

	High excludability	Low excludability
High subtractability	**Private goods** farmland, minerals, bottled water	**Common-pool resources (CPRs)** aquifers, forests, fisheries
Low subtractability	**Club goods** patents, toll roads, satellite TV	**Pure public goods** climate stability, national defence

- **Private goods:** if I eat an apple, you can't (high subtractability), and I can put it in my bag (high excludability).
- **Club goods:** my watching Netflix doesn't stop you (low subtractability), but no subscription means no access (high excludability).
- **Pure public goods:** low subtractability and low excludability, e.g., atmospheric stability.
- **CPRs: high subtractability, very low excludability**. Pumping groundwater removes it for everyone, but stopping 10,000 farmers from drilling private boreholes is nearly impossible without massive policing costs.

Four property regimes for a CPR

EXPLAINED SIMPLY

A resource of the same type can be run in four ways: nobody owns it (open access), the government owns it, a private party owns it, or a community owns it with its own rules.

Example: The same lake could be open to anyone, run by the fisheries department, owned by a private resort, or managed by a fishermen's co-operative.

The **nature of a good** is not the same as the **regime governing it**. A CPR can exist under any of four regimes:

1. **Open access:** no one owns it and no rules exist. *This is what Hardin actually modelled.*
2. **State property:** government claims ownership and sets laws.
3. **Private property:** individuals or corporations hold exclusive title.
4. **Common property:** an identifiable community owns it, with internal rules and external boundaries.

The bundle of rights (Schlager & Ostrom, 1992)

EXPLAINED SIMPLY

Ownership isn't all-or-nothing. It's a bundle of five rights, from simply entering, to taking resources, to making rules, to deciding who gets in, to selling it. Communities can manage resources well with the first four, without the right to sell.

Example: A housing society's residents can enter the garden (access), pluck flowers (withdrawal), set garden timings (management) and refuse outsiders (exclusion), but no one can sell the garden (alienation). It still stays well kept.

Ownership breaks down into five **cumulative** rights (each tier includes those below it):

Tier	Right	Meaning	Example
1	**Access**	Enter the physical space	Walk on the beach
2	**Withdrawal**	Harvest units	Fish from the beach
3	**Management**	Regulate use	Set minimum fish size, seasons
4	**Exclusion**	Decide who gets access and withdrawal rights	Village decides who may fish
5	**Alienation**	Sell, lease or transfer all lower rights	Sell the fishing rights

OSTROM'S REBUTTAL TO HARDIN

Hardin's assumption: unless someone holds Tier 5 (full private alienation), the resource collapses. **Ostrom's empirical finding:** groups holding only **Tiers 1–4** can govern resources sustainably **for hundreds of years**, without privatising or bringing in state police.

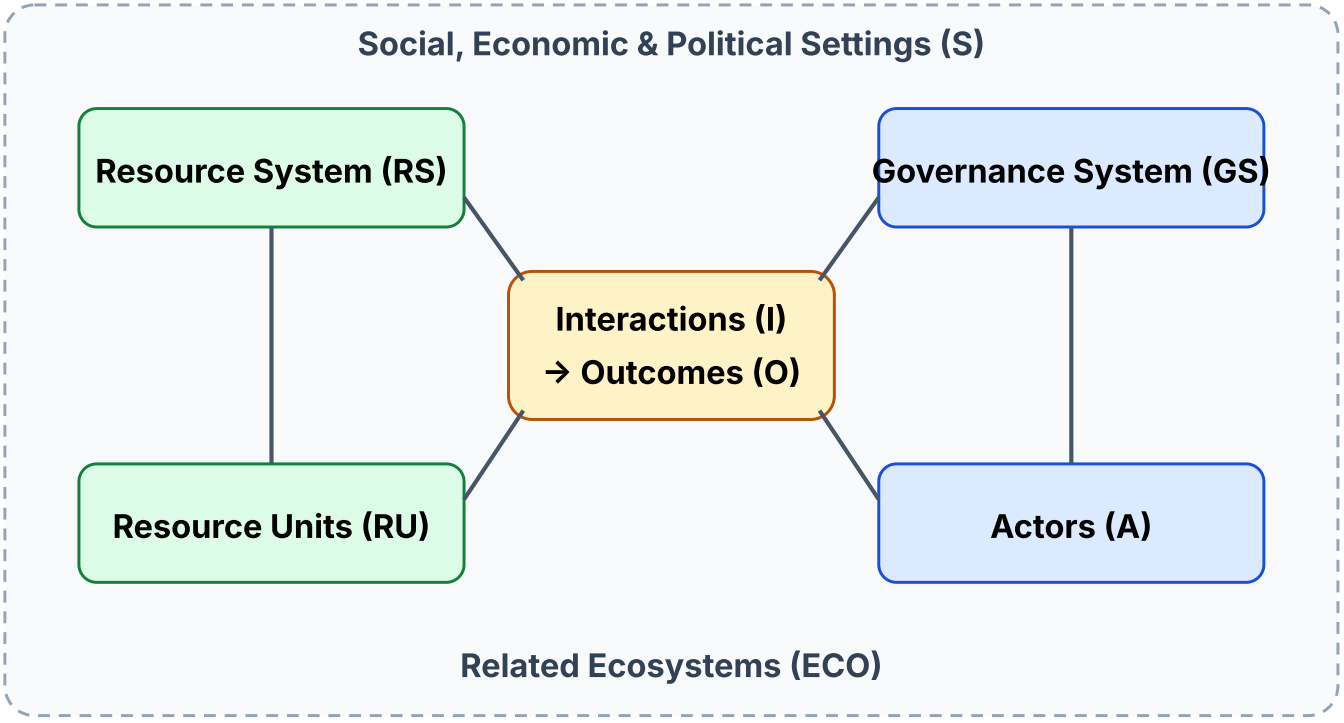
7.6 Elinor Ostrom and the SES framework

EXPLAINED SIMPLY

Ostrom said there's no one-size-fits-all fix. Look at four things: the resource (how big, how bounded), the units taken (fixed like trees or moving like fish), who makes the rules, and who the users are. When these fit together, the commons survives.

> **Example:** A small village pond with a local committee and families who depend on it fully is easy to manage. The open ocean, with migrating tuna, distant regulators and global fleets, is very hard.

In **2009**, **Elinor Ostrom** became the **first woman to win the Nobel Prize in Economic Sciences**. She showed that humans are neither prisoners of a fixed tragedy nor saints. Instead of panaceas ("privatise everything" or "nationalise everything"), she treated sustainability as a multi-tier diagnostic problem: the **Socio-Ecological Systems (SES) Framework**.



Ostrom's SES framework: four core sub-systems interact to produce outcomes, within wider social settings and related ecosystems.

Sub-system	Diagnostic question
Resource System (RS)	The overall ecosystem: a 50-hectare lake or the Pacific Ocean? Clearly bounded or infinite?
Resource Units (RU)	The extracted units: stationary like timber, or mobile and unpredictable like migratory tuna?
Governance System (GS)	Who makes the rules: an informal village council or a bureaucratic capital-city agency?
Actors (A)	The users: their social capital, and whether the resource is 100% of their livelihood or just a portfolio asset.

When RS, RU, GS and A **align**, you get sustainable interactions and resilient outcomes. When they **misalign**, the system collapses.

Empirical base: Ostrom studied thousands of cases, including **500-year-old communal irrigation systems in Spain** and **shared mountain pastures in Törbel, Switzerland**. Systems that lasted for centuries **never relied on pure markets or pure state force**. They followed **eight design principles**.

7.7 Ostrom's eight design principles

EXPLAINED SIMPLY

Ostrom found eight common features of shared resources that have lasted for centuries, such as clear boundaries, rules made by users, monitoring by users, mild punishments first, and quick local dispute resolution.

Example: Kerala's traditional fishing communities and Rajasthan's johad water-harvesting villages follow most of these principles without any private ownership or police.

#	Principle	What it means
1	**Clearly defined boundaries**	Clear physical boundaries around the resource, and precise rules on who has legitimate rights to use it.
2	**Congruence with local conditions**	Rules match local reality, not a rigid outside calendar.
3	**Collective-choice arrangements**	Those who must follow the rules have a direct voice in making them.
4	**Accountable monitoring**	Monitors are the users themselves, or directly accountable to them.
5	**Graduated sanctions**	Start with a light warning; reserve heavy penalties for habitual offenders.
6	**Rapid, low-cost conflict resolution**	Local arenas, not decades of litigation.
7	**Minimal recognition of rights to organise**	National, regional or municipal governments don't challenge, overturn or undermine local communities' authority.
8	**Nested enterprises**	For large systems (multi-basin watersheds, regional forests, coastal waters), governance is organised across multiple nested tiers.

MEMORY AID: "B-C-C-M-S-C-R-N"

Boundaries · **C**ongruence · **C**ollective choice · **M**onitoring · **S**anctions (graduated) · **C**onflict resolution · **R**ecognition of rights · **N**ested enterprises. Think of a housing society that runs its own water tank: clear membership, rules suited to the building, residents vote on rules, residents check the meter, a warning before a fine, the committee settles disputes, the municipality respects the society, and the society sits within the city's water system.

Notice what's missing: neither private titles nor state force. The linchpin is **organised community monitoring and reciprocal trust**.

7.8 Are humans really *Homo economicus*?

EXPLAINED SIMPLY

Old theory said people always cheat if they can. Ostrom found most people are 'conditional cooperators': they play fair if others do, and stop if they feel cheated. Building trust is therefore the key.

> **Example:** In an office tea fund, people contribute happily when everyone pays. Once someone is seen skipping payments, others start skipping too. Visible honesty keeps it going.

- **Mancur Olson, *The Logic of Collective Action* (1965):** rational individuals will always free-ride unless forced. Hardin agreed.
- **Ostrom's behavioural finding:** humans are **conditional cooperators**. We cooperate if we believe others will, and we defect if we think we're being played for a sucker.

THE TRUST–RECIPROCITY–REPUTATION TRIAD

When monitoring is visible and sanctions graduated, the risk of being cheated drops, **trust** rises and **reciprocity** stabilises, all supported by **reputation**. The tragedy is averted **without authoritarian intervention**.

7.9 Game theory of the commons: Hardin vs. Ostrom

EXPLAINED SIMPLY

Without punishment, cheating pays more whatever the other person does, so both cheat and both lose. Add a fine for cheating, and honesty becomes the better choice for everyone.

Example: In an exam with no invigilator, copying looks tempting. With an invigilator and a penalty, honest work becomes the smart choice.

NUMERICAL 7 – SOLVING BOTH GAMES (PAYOFFS WRITTEN AS (ROW PLAYER, COLUMN PLAYER))

Game 1: Hardin's commons (one-shot, no sanctions)

	Cooperate	Defect
Cooperate	(3, 3)	(0, 5)
Defect	(5, 0)	(1, 1)*

Game 2: Ostrom's commons (monitoring + sanction S = 3)

	Cooperate	Defect
Cooperate	(3, 3)*	(0, 2)
Defect	(2, 0)	(−2, −2)

"Cooperate" = keep 10 cows (restraint); "Defect" = add cows (over-use).

Solving Game 1 (Hardin)

1. **If the other player Cooperates:** I get 3 by cooperating or 5 by defecting. 5 > 3, so **Defect**.
2. **If the other player Defects:** I get 0 by cooperating or 1 by defecting. 1 > 0, so **Defect**.
3. Defect is best whatever the other does, so **Defect is the dominant strategy**. By symmetry the other player defects too.
4. **Nash equilibrium = (Defect, Defect) = (1, 1)**. Total welfare = 2, vs 6 if both cooperated. This is a **prisoner's dilemma** that leads to collective collapse.

Building Game 2: apply the sanction (the professor's method)

1. Monitoring catches every defector, who pays a sanction **S = 3**. Subtract 3 from the defector's payoff only; cooperators' payoffs don't change:

```
Defect vs Cooperate: 5 − 3 = 2      → cell (D,C) becomes (2, 0)
Defect vs Defect:    1 − 3 = −2     → cell (D,D) becomes (−2, −2)
Cooperate vs Defect: (0, 5 − 3)     → cell (C,D) becomes (0, 2)
Cooperate vs Cooperate: unchanged  → (3, 3)
```

Solving Game 2 (Ostrom)

2. **If the other Cooperates:** cooperate gives 3, defect gives 2. 3 > 2, so **Cooperate**.

3. **If the other Defects:** cooperate gives 0, defect gives −2. 0 > −2, so **Cooperate**.
4. Cooperate is now the **dominant strategy**. **Nash equilibrium = (Cooperate, Cooperate) = (3, 3)**, with total welfare 6, the best possible.

Extension: what is the minimum sanction that works?

1. For Cooperate to beat Defect when the other cooperates: 3 > 5 − S, so **S > 2**.
2. For Cooperate to beat Defect when the other defects: 0 > 1 − S, so **S > 1**.
3. Both must hold, so the sanction must be **greater than 2**. The class value S = 3 satisfies this. A sanction of 1.5 would not stop defection when others cooperate.

Hardin: Nash (Defect, Defect) = (1, 1), collapse. Ostrom with S = 3: the payoffs shift (5 − 3 = 2; 1 − 3 = −2), and (Cooperate, Cooperate) = (3, 3) becomes the stable equilibrium. Any sanction S > 2 achieves this.

7.10 Applied case: corporate water stewardship

EXPLAINED SIMPLY

Instead of grabbing water through permits or land purchases (Hardin's way), the company joins farmers and the city to measure, share and jointly manage the aquifer (Ostrom's way).

Example: A beverage plant co-funds water meters on all major borewells, shows the data publicly, and pays into a recharge fund whenever it exceeds its seasonal limit.

Scenario: you are CSO of a multinational beverage firm whose bottling plant sits on a rapidly depleting aquifer in an arid basin.

Hardin's playbook (two options)	Applying Ostrom's SES framework instead
1. Lobby the state for a monopoly permit and enforcement against local farmers. 2. Buy surrounding farmland to claim absolute private extraction rights. **Result:** communities riot, your social licence vanishes, brand equity plummets, and farmers drill illegal wells deeper than yours anyway.	**Boundaries:** partner with the regional watershed board to map hydrological limits. **Collective choice:** form a water users' association of industry, municipal utilities and farmer cooperatives. **Monitoring:** co-fund real-time telemetry on major wells, shown on a public dashboard. **Graduated sanctions:** exceed your seasonal drawdown threshold and you pay tiered tariffs into a local aquifer replenishment fund.

LINK TO PLACHIMADA (MODULE 5)

Coca-Cola's Plachimada plant followed the "Hardin playbook": it relied on state permits and private land rights, and it lost everything. The Ostrom approach is exactly what the Class 2 capstone recommends.

7.11 Polycentric governance and closing synthesis

EXPLAINED SIMPLY

Polycentric means 'many centres of decision-making' at different levels (village, state, national, global) working together under common rules, rather than one boss for everything.

Example: Water in India is managed by Panchayats, state boards, central ministries and river-basin bodies all at once. Each handles its own level, coordinating under shared policies.

POLYCENTRIC GOVERNANCE

Multiple independent centres of authority, at different scales, coordinating under common rules. Instead of waiting for a single global regulator or retreating behind private-property walls, businesses embed themselves in durable local collective institutions.

Where Hardin was right	Where Hardin was wrong
The sustainability problem has **no purely technological fix**. A world run on solar panels and carbon capture collapses just as fast as one run on coal if governance rewards unconstrained extraction.	He assumed people in a commons are passive victims of game theory who can't talk to each other. **Ostrom proved** that with clear boundaries, legal space to organise and mechanisms for trust, communities can govern the commons sustainably across generations.

MODULE 8

Quick Revision

Formula sheet · all numericals at a glance · glossary · how the topics connect

8.1 Formula sheet

Concept	Formula / rule
Weak sustainability (Hartwick–Solow)	$d/dt\ (K_m + K_f + K_h + K_n) \geq 0$
Strong sustainability	$K_n \geq K_{crit}$ (critical natural capital must be kept; no substitution)
Efficient allocation	MB = MC; total surplus = CS + PS maximised
Market failure condition	MSC ≠ MSB at market output
Negative production externality	MSC = MPC + MEC → $Q^* < Q_m$ → Pigouvian tax $t^* = MEC(Q^*)$
Negative consumption externality	MSB = MPB − MEC → $Q^* < Q_m$ → tax or regulate consumption
Positive production externality	MSC = MPC − MEB → $Q^* > Q_m$ → subsidy $s^* = MEB(Q^*)$
Positive consumption externality	MSB = MPB + MEB → $Q^* > Q_m$ → subsidy or public provision
Deadweight loss (linear curves)	$DWL = ½ \times (Q_m - Q^*) \times MEC$ (or MEB)
CBAM fee	Fee = emissions intensity (t CO_2/t product) × (EU carbon price − home carbon price paid)
Landed cost in EU	Base production cost + CBAM fee
PES feasibility	$WTP_{beneficiary} \geq$ Payment $\geq WTA_{provider}$ (WTA = opportunity cost)
PES bargaining zone (class game)	$\Pi_{Conserve} + T > \Pi_{Clear}$ and $\Pi_{Clean} - T > \Pi_{Plant} \rightarrow 70 < T < 140$
Welfare and DWL in games	W = sum of both payoffs; $DWL = W_{optimum} - W_{Nash}$
Hardin's herdsman	Net utility = +1 − 1/n (> 0 for any n > 1)
Sanction game	Defector's payoff − S; cooperation dominant if S > 2 (class numbers)
Water balance	Ending stock = Beginning stock + (Inflows − Outflows)
DSCR	Cash available for debt service ÷ debt service due

8.2 All numericals at a glance

#	Problem	Key answer
1	CBAM steel case (Module 3)	Fees: A $189, B $27 → landed A $769, B $777; A's $170 edge falls to $8. At $110: A $811 > B $783. Break-even ≈ $94.4/t CO_2.
2	Pigouvian tax practice (Module 4)	Q_m = 40, Q* = 30, DWL = 100, t* = 20.
3	Pigouvian subsidy practice (Module 4)	Q_m = 40, Q* = 45, DWL = 25, s* = 10.
4	Uniform cap vs tax (Module 4)	$3,000 vs $1,000 for the same 100 t cut.
5	PES upstream–downstream game (Module 6)	Nash (Clear, Plant) W = 40; optimum W = 110; DWL = 70; zone 70 < T < 140; at T = 80: (110, 0).
6	Hardin's herdsman (Module 7)	Net utility = 1 − 1/10 = +0.9, so add the cow.
7	Hardin vs Ostrom games (Module 7)	Hardin: (D, D) = (1, 1). Ostrom with S = 3: (C, C) = (3, 3). Need S > 2.
–	Weak-sustainability rule check (Module 3)	ΔTotal K = −20, so it fails even weak sustainability.
–	Case figures (Modules 1, 4, 5, 6)	AHS ≈ 19% CAGR; LSE overrun ≈ 67×; Flint 120× the "saving"; Plachimada = water for 10–20k people; Vittel €1,400/ha.

8.3 How the topics connect

THE STORY OF THE COURSE IN SIX SENTENCES

1. **Nature is a form of capital**, and part of it (K_{crit}) can't be replaced: strong over weak sustainability.
2. The economy sits **inside** society, which sits inside the biosphere, so E comes first in ESG.
3. Markets fail to protect nature because of **externalities, the horizon problem and information asymmetry**.
4. Fixes include **Pigouvian taxes/subsidies, regulation, disclosure (CSRD/ISSB), border taxes (CBAM)** and **Coasean deals such as PES**.
5. But global compliance isn't enough: water and land are **local, non-fungible commons**, so firms need a **social licence** (Plachimada, Flint, Cauvery).
6. **Ostrom** shows communities can govern commons through trust, monitoring and graduated sanctions, and smart firms join these polycentric institutions. Strategically, firms should move from Defender to **Winner** (Lubin & Esty).

8.4 Glossary

Term	Meaning in plain words
Additionality	Would the conservation have happened anyway without payment? If yes, the payment adds nothing.
Annex I	Developed countries with binding targets under Kyoto.
CBAM	EU Carbon Border Adjustment Mechanism: a carbon charge on imports based on their embedded emissions.
CBDR	Common But Differentiated Responsibilities: everyone acts, but rich countries lead.
CDM	Clean Development Mechanism: Kyoto tool for earning credits from projects in developing countries.
Carbon leakage	Polluting production moves to places with no carbon price.
Coasean bargaining	With clear rights and low transaction costs, parties can negotiate an efficient fix themselves.
Conditionality	Pay only if the service is actually delivered.
CPR	Common-pool resource: rival but hard to exclude (aquifers, fisheries, forests).
CSRD / ISSB	EU reporting directive (double materiality) / global standards board (financial materiality).
CSO	Chief Sustainability Officer.
Deadweight loss	Welfare lost because the wrong quantity is produced; nobody gains it.
Double materiality	Reporting both how the world affects the firm (financial) and how the firm affects the world (impact).
DSCR	Debt-Service Coverage Ratio: cash available ÷ debt payments due.
Eco-premium	Extra shareholder value from strong sustainability execution (lower risk, lower cost of capital).
Externality	A side-effect on third parties that isn't reflected in the price.
Fungible	Interchangeable: one unit is as good as another anywhere (carbon is; water is not).
Greenwashing	Looking green through PR without real change.
Harmon Doctrine	The (rejected) idea that an upstream country can do anything with "its" river.
IWRM	Integrated Water Resources Management: plan by river basin, not by state borders.
K_{crit}	Critical natural capital: the minimum nature we cannot replace or lose.
Leakage (PES)	Protection in one place pushes damage to another place.
MEC / MEB	Marginal external cost / benefit: the extra harm or help to others from one more unit.
Moral hazard	Behaving more carelessly because someone else bears the cost.

NDC	Nationally Determined Contribution: each country's own Paris pledge.
NRDA	Natural Resource Damage Assessment: valuing nature's losses after a disaster.
Pareto efficient	No one can be made better off without making someone else worse off.
Pigouvian tax	A tax equal to the external damage per unit.
Planetary boundaries	Nine Earth-system limits; six are already breached.
Polycentric governance	Many centres of authority at different scales coordinating under common rules.
Public Trust Doctrine	The state holds natural resources in trust for the people and is not their owner.
Social licence to operate	The community's acceptance of a business, beyond its legal permit.
Tragedy of the horizon	Climate damage arrives after the decision-makers' time horizon ends (Carney).
WTP / WTA	Willingness to pay (buyer's maximum) / willingness to accept (seller's minimum).
WUA	Water Users' Association: farmers acting collectively as a "seller" of water.

Note: problems marked "practice" or "illustrative" use invented numbers to show the professor's method. All figures from the class documents are reproduced as given.